Garbling a Hash-Based STARK Verifier for Bitcoin

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Abstract

BitVM-style protocols verify succinct proofs on Bitcoin by garbling a proof verifier off-chain. Existing garbled-verifier and witness-encryption designs check elliptic-curve SNARKs, which do not resist a quantum adversary. A hash-based STARK verifier removes that dependency but is difficult to garble: it branches on data, its transcript is defined by an implementation, hashing and field multiplications dominate its cost, and a stored circuit of 10^8 gates does not fit in memory. We garble the Plonky3 verifier path exercised by a Keccak-f algebraic intermediate representation (AIR) without public values over WHIR, using the binary tower field $\mathbb{F}_{2^{128}}$ and Blake3 for commitments, the transcript and garbling. Additions are free under free-XOR, a multiplication costs 2,187 AND gates, a Blake3 compression costs 10,281, and a streaming garbler holds one byte per wire and one label per live wire. For a 2^{18}-row trace at 104 bits of STARK soundness, the verifier has 94.1M non-free gates and garbles to 1.5 GB in under a minute on one core, about 30× fewer non-free gates than the Boolean-garbled Groth16 verifier. We then give a conditional BitVM3-style integration and isolate its largest measured Bitcoin-side cost. The circuit consumes 1,041,024 input bits (127.1 KiB), distinct from the 140 KiB serialized proof. A policy-shaped Taproot fixture carries the raw bitstream in 33,299 virtual bytes (vB), but it has no signature or garbled-label authentication and is not a protocol cost. A separate Core-tested P2TR envelope authenticates the 130,128-byte payload in 33,004 vB, but neither delivers the selected labels nor includes its data-availability and fraud graph. The accounted input-authentication component, including the adaptor or one-time-signature witness and its verification script, measures 2.22 million vB (MvB) for the 8-bit Schnorr-adaptor baseline or 11.85–17.27 MvB for hash-based keys. Under strict Bitcoin Core 31.1 policy, all three signed reveal sets were accepted after their funding transactions confirmed. Funding plus reveals totals 2.228, 11.883 and 17.311 MvB, respectively; a tested Lamport join raises its bounded slice to 17.324 MvB. The three mechanisms still do not have identical transaction-shell boundaries, and no complete dispute graph has been constructed; they therefore cannot substitute for a measured total on-chain cost. The hash-based figures also exclude the unimplemented soldering from one key set to the seven kept garblings; the arithmetic projection for seven independent key sets, whose consistency protocol is also unimplemented, is 82.94–120.86 MvB before the missing graph overhead. The measured proof system provides 104 classical bits and about 52 quantum bits, whereas the reported cut-and-choose parameters provide about 40 classical bits and Bitcoin's transaction signatures are not post-quantum. Thus the artifact is a measured hash-based verifier and a cost study, not a complete post-quantum Bitcoin protocol. Recursing a zkVM proof through a narrow tape machine, with experimental and unreviewed changes to the proof system that we list, brings the garbled verifier's input to 564,552 bits in 218.0M non-free gates. Each of its proofs meets 100 bits by Plonky3's assessment, which does not count those changes, and the circuit binds neither the zkVM program nor its outputs.

1 Introduction

Bitcoin cannot verify a succinct proof economically within current Script and policy limits, but a trust-minimized bridge needs such verification. Consider a withdrawal from a rollup: an operator has paid out a user on Bitcoin and must prove, to a Bitcoin script, that the rollup's state authorizes

the payment; the proof is a STARK over the rollup's execution. This bridge is the motivating application. The implementation evaluated in this paper verifies a Keccak-f AIR without public values, so it does not yet bind a proof to a withdrawal. Hashing dominates such statements, and we evaluate on a hash-heavy standard case, proofs of Keccak-f permutations [Nat15].

A SNARK verifier written in Script takes hundreds of megabytes [GKS+26]. The BitVM line of work therefore verifies *optimistically*: a prover claims a result, and a challenger disputes it only if it is wrong. BitVM2 [LAA+25] settles a dispute by bisecting the verifier's execution trace on-chain, which is secure but expensive: a mainnet experiment put the unhappy path at over \$14,000 [GKS+26]. BitVM3 moves the verifier off-chain into a *garbled circuit* [Lin25, WAA+26, Che25]. At setup the verifier circuit is garbled. At dispute time the proof is published under one-time signatures that give the evaluator the circuit's input labels, and the outcome depends on which output label the evaluation yields. For the Groth16-sized instances reported by BitVM3 and BABE, only the signed proof and a hashlock remain on-chain, for tens of dollars [GKS+26].

The off-chain side is where the existing designs fall short. A garbled circuit has to be generated, stored for the lifetime of the deposit and checked, because the garbler may cheat, typically by cut-and-choose over many instances. The Boolean-garbled Groth16 verifier [Bit25] has about 3×10^9 non-free gates, some 42 GiB, and hours of cut-and-choose setup [GKS+26]. Garbling Groth16 with native group operations [KFLT26] and witness encryption [GKS+26] shrink the artifact to megabytes. All of these designs, however, verify an elliptic-curve SNARK, pairing-based or its designated-verifier form over BN254 or a binary curve: a quantum adversary breaks them, and they need a trusted or secret setup. A STARK proof must first be wrapped in such a SNARK, which reintroduces both.

Key idea. We garble a hash-based STARK verifier directly and make it cheap by choosing its field. Under free-XOR [KS08] a garbled circuit pays only for non-linear gates, and in a field of characteristic two every addition is an XOR. A STARK over the binary tower field $\mathbb{F}_{2^{128}}$ with Blake3 commitments therefore costs, as a garbled circuit, its hashes and field multiplications, with additions free. Its algebraic checks rest on WHIR over $\mathbb{F}_{2^{128}}$, while Blake3 instantiates the commitments, the Fiat–Shamir transcript and the garbling hash, modeled as circular correlation robust (§5).

Challenges. Four problems stand between this idea and a working garbled verifier.

- C1 *A fixed circuit for a branching verifier.* A garbled circuit has one shape for all inputs, while the verifier redraws challenges and fixes coordinates depending on sampled values.
- C2 *A transcript defined only by code.* The circuit must absorb and sample exactly the bytes that the prover's implementation does, or Fiat–Shamir soundness is lost, and that byte schedule is specified nowhere outside the implementation.
- C3 *Hashing is expensive.* Merkle paths, leaf rows and the transcript are Blake3 compressions, and a compression costs more than four field multiplications.
- C4 *Memory.* A verifier of 10^8 non-free gates has several hundred million wires; a builder that stores the gate list needs tens of gigabytes before garbling starts.

Contributions. The first three contributions answer C1–C4; the last three study its Bitcoin integration, security boundaries and measurements.

- **A fixed-shape circuit for the measured verifier path (C1, C2).** We implement the path through Plonky3's multi-STARK verifier [Plo26] exercised by our Keccak-f AIR over the Boolean WHIR trace commitment as a Boolean circuit with one output wire: the transcript, the zerocheck of the AIR, the batching of the opened columns, the bit ring switch and the WHIR opening. Data-dependent branches are fixed and guarded, and the transcript follows a byte schedule recovered from Plonky3's own verifier, so one circuit serves every proof of a configuration and can be garbled before the proof exists (§3).
- **Cheaper gadgets (C3).** Tower-field multiplication at 3^k AND gates per 2^k bits, with squaring and multiplication by the tower generator free, and a Blake3 circuit at 10,281 AND gates per compression instead of 20,657, with tree mode for long inputs (§3.1).
- **A streaming garbler (C4).** A plan pass counts each wire's uses, and the garbling pass garbles and checks each gate as the builder emits it and releases a wire's slot after its last use (§4).
- **A Bitcoin integration study and explicit deployment gaps.** We price three ways to publish the proof, sweep the proof parameters, and separate the accounted one-key-set input-authentication component from the unimplemented multi-copy soldering and transaction graph (§6). Including signature or preimage witness data and authentication scripts, this component measures 2.22 MvB with the 8-bit adaptor baseline and 11.85–17.27 MvB per hash-key set, against a few kilobytes for Groth16. These are not total on-chain footprints. Tuning removes only a fifth of the logical input.
- **A scoped quantum analysis.** We analyze the STARK and garbling components in the quantum random-oracle model, including Grover search on the garbling offset, and separate those bounds from the lower cut-and-choose level and the non-post-quantum Bitcoin transaction layer (§5.1).
- **Measurements.** Real Keccak-f proofs from 2^5 to 2^{20} rows, a gate profile, an ablation of the gadget and memory choices, measurements of the one-key-set input-authentication components, seven-key-set projections, a comparison with pairing-based designs, and the garbled verifier of a recursion proof of Ziren's binary stage at 564,552 input bits (§7).

The circuit builder and the measurements are open source; the streaming garbler, the Blake3 circuit and a label-level evaluator are contributed to GOAT's bitvm-gc [GOA26b].

2 Setting

2.1 Garbled verifiers on Bitcoin

We follow the BitVM3 setting [WAA+26]. A *garbler*, the operator who proves, and an *evaluator*, a challenger, agree on a verifier circuit C with one output wire, and C is garbled at setup. At dispute time the operator publishes its proof on-chain under one-time keys, and each published bit gives the evaluator the input label of that bit's value. The proof is public, so we use *privacy-free* garbling: the evaluator may learn every wire's value. What garbling must provide is *authenticity*. Holding the labels of an input x, the evaluator obtains the label of the output value $C(x)$, and it cannot obtain the label of the other value. A hashlock on the label of value 0 lets a challenger who evaluates an invalid proof stop the operator's claim.

The garbler may cheat, so the garbling is checked by cut-and-choose, as in BABE [GKS+26], Mosaic [KTA+26] and bitvm-gc [GOA26b]; its cost is linear in the size of the garbled circuit, which this paper reduces. Section 6 gives an integration sketch, accounted input-publication components and the unresolved deployment requirements.

2.2 The garbling scheme

We use privacy-free garbling [FNO15] with free-XOR [KS08] and one ciphertext per non-free gate, in the syntax of Bellare, Hoang and Rogaway [BHR12b].

Let Δ be a random secret 128-bit offset. Every wire w has a false label w_0, and its true label is $w_1 = w_0 \oplus \Delta$. Let $H(\ell, g) = \text{Blake3}(\ell \,\|\, g)$ truncated to 128 bits, where g is the gate's index. An XOR gate sets $c_0 = a_0 \oplus b_0$ and has no ciphertext. An AND gate $c = a \wedge b$ sets

$$c_0 = H(a_0, g), \qquad ct = H(a_1, g) \oplus H(a_0, g) \oplus b_0.$$

An evaluator holding the label a of value x, and b of value y, outputs $H(a, g)$ if $x = 0$ and $H(a, g) \oplus ct \oplus b$ if $x = 1$. In both cases this is the label of $x \wedge y$. An OR gate is symmetric, with $c_0 = H(a_1, g) \oplus \Delta$ and $ct = H(a_1, g) \oplus H(a_0, g) \oplus b_1$. The garbled circuit is one 16-byte ciphertext per AND or OR gate.

Two details matter in practice. First, Δ must be secret and fresh for each garbling: with a public Δ any label yields its complement, and the output's other label is free to compute. Second, a hash modeled as circular correlation robust (§5) gives labels no algebraic structure. RSA-based garbling proposals for BitVM3 were broken exactly through such structure: label forgery by extended Euclid on coprime exponents, and by the Franklin–Reiter related-message attack [FLB25].

2.3 The proof system

We target Plonky3's multi-STARK verification pipeline [Plo26] over the Boolean WHIR trace commitment. The evaluated implementation instantiates one AIR, with no public values, preprocessed columns or lookups, and requires current- and next-row openings for every column. The pipeline uses fields in the Wiedemann tower [Wie88]

$$\mathsf{T}_0 = \mathbb{F}_2, \qquad \mathsf{T}_{k+1} = \mathsf{T}_k[X_k]/(X_k^2 + X_{k-1}X_k + 1), \qquad X_{-1} = 1,$$

up to $\mathsf{T}_7 = \mathbb{F}_{2^{128}}$. An element of T_{k+1} is stored as its low half (the coefficient of 1) and its high half (the coefficient of X_k), recursively down to bits. The trace is committed as bits: a Boolean column is packed into $\mathbb{F}_{2^{128}}$ elements, and WHIR commits to the packed multilinear polynomial on an additive (Cantor) evaluation domain [Can89, LCH14]. Binary towers and ring switching come from Binius [DP25, DP26].

Terms. An *AIR* (algebraic intermediate representation) is a set of polynomial constraints over a trace's current and next rows. A *zerocheck* [CBBZ23] proves that the constraints vanish on every row by a sumcheck [LFKN92] against the equality polynomial $\text{eq}(\tau, \cdot)$ at a random point τ. *WHIR* [ACFY25] is a Reed–Solomon proximity test used as a multilinear polynomial commitment. Its *STIR queries* [ACFY24] open the committed, folded codeword at random positions, and its *out-of-domain* (OOD) samples bind the committed polynomial at points outside the evaluation domain. The *Cantor basis* is the $\mathbb{F}_2$-basis $v_0 = 1$, $v_i^2 + v_i = v_{i-1}$, which gives binary fields an additive FFT domain. The *bit ring switch* turns a claim about a polynomial's bits into a claim about its packed $\mathbb{F}_{2^{128}}$ representation. *Johnson list-decoding regime* refers to the proximity parameters for which the soundness bounds are proven via the Johnson bound, without conjectures.

Verification. The verifier:

1. runs a *zerocheck* of the AIR: a degree-4 sumcheck over the row variables, then the AIR constraints, evaluated at the bound point from the opened current and next rows and Horner-batched under a challenge α, against $\mathrm{eq}(\tau, r)$;
2. batches the opened columns into one bit-level claim, at a column point drawn from the transcript;
3. runs the *bit ring switch*, which turns that claim into an $\mathbb{F}_{2^{128}}$ claim on the packed polynomial. It uses a 128×128 tensor and, above 2^7 rows, its carry and last tensors, a batched sumcheck, and a closing weight computed in the tensor algebra;
4. runs the WHIR opening of that claim: sumcheck rounds with proof of work, STIR queries on Merkle-committed folded codewords, and a final polynomial.

All challenges come from a Blake3 [OANWO20] sponge over bytes, and all Merkle trees use Blake3 compressions.

3 The verifier as a circuit

This section turns the verifier of §2 into a Boolean circuit with one output wire. Figure 1 shows its stages in transcript order, with each stage's share of the non-free gates of our running example: a 2^{18}-row Keccak-f trace with 1,625 columns, 1,650 AIR constraints and 22 packed variables, whose proof is 140 KiB and whose verifier has 94.1M non-free gates. We first describe the gadgets, then the transcript, the multi-STARK layers and WHIR.

3.1 Gadgets

Tower arithmetic. A wire vector for an element of T_k is its bit pattern, so gadget outputs can be read back directly as field elements. Multiplication follows the tower recursively, Karatsuba at each level:

$$z_0 = a_0 b_0, \quad z_2 = a_1 b_1, \quad z_1 = (a_0 + a_1)(b_0 + b_1) + z_0 + z_2, \qquad ab = (z_0 + z_2) + (z_1 + X_{k-1} z_2)\, X_k.$$

Additions are free. Multiplication by the generator X_{k-1} is a rewiring plus XORs, and squaring is linear over $\mathbb{F}_2$. The only AND gates are the base-field products, three per level, so a 2^k-bit multiplication costs 3^k AND gates: 2,187 in $\mathbb{F}_{2^{128}}$. A multiplication by a constant folds to XORs entirely.

Blake3. We start from bitvm-gc's Blake3 circuit and change two things. A 32-bit addition's carry is the majority

$$\mathrm{maj}(a, b, c) = c \oplus \big((a \oplus c) \wedge (b \oplus c)\big),$$

one AND per bit, and the carry out of the top bit is discarded, so it is not computed. This halves a compression from 20,657 to 10,281 AND gates. We also add Blake3's tree mode (chunk chaining values on a stack, parent nodes, the root last). The verifier's transcript absorbs more than one 1,024-byte chunk between samples: the final polynomial is 64 coefficients, 1,024 bytes on its own. Table 1 gives the costs. Every gadget is checked against the reference implementation (`p3-binary-field`, the `blake3` crate) on random inputs.

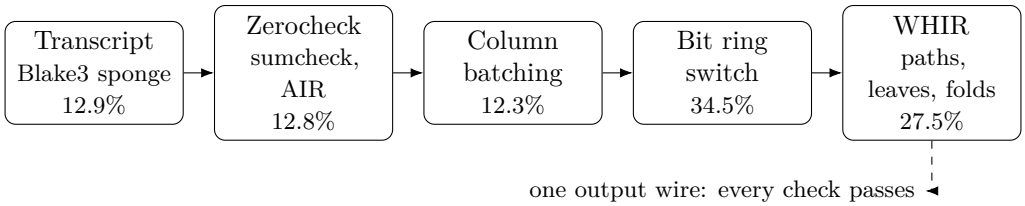


Figure 1: The ring switch dominates the verifier circuit, followed by WHIR. The stages appear in transcript order, with each stage's share of the 94.1M non-free gates of the 2^{18}-row verifier: transcript is the STARK and WHIR sponges, zerocheck is its sumcheck, AIR constraints and closing, and WHIR is everything after the ring switch except its transcript.

gadget	AND gates
$\mathbb{F}_{2^{128}}$ multiplication	2,187
$\mathbb{F}_{2^{128}}$ squaring, multiplication by X_{k-1}, addition	0
Blake3 of 64 bytes (a Merkle compression)	10,281
Blake3 of 256 bytes (a leaf row of 16 elements)	41,511
Blake3 of 1,072 bytes (two chunks: a transcript flush)	186,125

Table 1: Gadget costs, checked against `p3-binary-field` and the `blake3` crate.

3.2 Transcript

The transcript is Plonky3's challenger over a Blake3 byte sponge. Observing appends to an input buffer; sampling, when the output buffer is empty, hashes the input buffer, replaces it with the digest, and pops bytes from the digest's end. A field element is 16 little-endian bytes. A query index is 8 bytes read as a little-endian integer and masked. A proof-of-work check absorbs the witness and requires a fixed number of sampled bits to be zero. We implement the same challenger on wires. Its schedule has no data-dependent branch: queries are a fixed number of masked draws, and a proof-of-work check a fixed number of bits. So the circuit's shape is a function of the configuration alone. The sub-transcripts' domain separators are constants, baked in as constant wires, and everything the prover sends is an input.

The byte schedule, that is, the order in which roots, out-of-domain answers, sumcheck messages, proof-of-work witnesses, stratified query draws and the final polynomial are absorbed and sampled, is implicit in Plonky3's implementation and not specified separately. We recovered it by running Plonky3's own verifier through a challenger that logs every byte, and we test the circuit against that log op for op.

3.3 The multi-STARK layers

The zerocheck, the column batching and the ring switch are implemented gate for gate from a reference implementation that is checked op for op against Plonky3's verifier. The AIR arrives as Plonky3's symbolic constraints and is evaluated on wires.

Two of Plonky3's data-dependent branches cannot appear in a fixed circuit. We fix them and guard them with checks.

- Plonky3 redraws a zerocheck challenge coordinate that happens to be zero. The circuit takes a fixed number of draws and requires each to be nonzero.

- Plonky3's ring switch would fix a column-point coordinate that is exactly 0 or 1. The circuit assumes there is no such coordinate and checks it.

Each event has probability about 2^{-127} per coordinate. When it occurs the circuit rejects where Plonky3 would continue, never the reverse.

3.4 WHIR

The WHIR layer checks every proof-of-work witness. It hashes every opened row and walks it up its Merkle path to the root absorbed in the transcript. It then folds each queried row with the round's folding randomness, and checks the final queries against the final polynomial and the closing sumcheck identity. Plonky3 sends Merkle openings pruned: siblings that another query can recompute are left out. Our implementation expands the proof outside the circuit into one full path per query. This is an engineering choice rather than an inherent limitation of a fixed circuit: a fixed upper-bound circuit could verify a padded frontier with data-dependent routing. A million-trial simulation of the actual stratified queries and cap forest, however, reduces the expected path from 1,354 to 1,335.71 digests, only 4,683 input bits or 0.45% of the total, before the routing and padding overhead. When the commitment uses a Merkle cap of 2^h roots, the transcript absorbs all of them, and each query selects its root with a multiplexer on the top h index bits: $256(2^h - 1)$ AND gates, against h compressions saved.

4 Streaming garbling

The circuits here have 10^8–10^9 non-free gates and several times as many wires. A builder that stores the gate list, as bitvm-gc's `CircuitAdapter` does, needs tens of gigabytes before garbling starts. Upstream's own DV-SNARK verifier test, for example, exceeds 64 GB. Nothing in garbling needs the gate list, though: a gate's ciphertext depends only on its input labels.

Our garbler is a circuit-builder backend that garbles each gate as it is emitted. It uses two passes of the same deterministic builder (Algorithm 1). The *plan* pass records, for each wire by creation index, the number of gates that read it: one byte per wire, with 255 meaning "never release". The *garbling* pass hands out slots. A wire's slot is released after its last read, and a later wire takes it. Wire identifiers are slots, which the builder interface permits. The two constant wires, and any wire read after the build (an output), are pinned. Both passes apply the same constant folding ($x \oplus x = 0$, $x \wedge 1 = x$, ...), so they see the same wires. Slot storage is chunked, so that a growing array never reallocates.

When the garbler holds a witness, it evaluates as it garbles. At each non-free gate it runs the evaluator's formula on the labels of the actual values, and requires the result to equal the label of the gate's output value. At setup no proof exists, and none is needed: garbling a gate uses only its input labels, so the garbler runs the garbling pass with every input held at 0 and keeps the ciphertexts. A challenger later evaluates them with an evaluator backend that runs the same builder on the proof and holds one label per wire. We tested exactly this. Garbled with every input at 0 and then evaluated, the stored garbling gave the label of 1 on a real proof and the label of 0 on the same proof with one opened value changed, at 2^5 and 2^8 rows. Evaluation took 11.5 s and 17.0 s, against 20.6 s and 34.0 s for garbling, since it hashes once per non-free gate where garbling hashes twice. Memory is one byte per wire for the plan, 0.61 GiB for the 659M wires at 2^{18} rows, plus 18 bytes per live wire (a 16-byte label, the value and a use counter), 41 MB at the peak of 2.26M live wires. The plan depends only on the circuit, so every garbled instance of a configuration can share it.

Algorithm 1 Streaming garbling of a deterministic builder $\mathcal{B}$

plan: run $\mathcal{B}$ on a backend that only counts: $uses[k] \leftarrow$ reads of the k-th created wire
pin the constants and the outputs: $uses[k] \leftarrow \infty$
garble: $\Delta \leftarrow \{0,1\}^{128}$; run $\mathcal{B}$ again on the backend:
  on input wire k: $\ell_0[s] \leftarrow \text{Blake3}_{seed}(k)$ in a free slot s; $v[s] \leftarrow$ its value
  on $\mathsf{XOR}(a, b)$: $\ell_0[s] \leftarrow \ell_0[a] \oplus \ell_0[b]$; $v[s] \leftarrow v[a] \oplus v[b]$
  on $\mathsf{AND}(a, b)$ with gate index g:
    $(c_0, ct) \leftarrow \text{Garble}(\ell_0[a], \ell_0[b], g, \Delta)$; emit ct
    $e \leftarrow \text{Eval}(\ell_{v[a]}[a], \ell_{v[b]}[b], ct, g, v[a])$
    $\ell_0[s] \leftarrow c_0$; $v[s] \leftarrow v[a] \wedge v[b]$; **assert** $e = \ell_{v[s]}[s]$
  on $\mathsf{OR}(a, b)$: as for AND, with the OR garbling of §2.2
  after each read of a: decrement $uses[a]$; release a's slot at zero

5 Security

This section states what the garbled verifier guarantees and on which assumptions. It combines the soundness of the proof system, the effect of the two guarded branches, and the authenticity of the garbling.

The circuit and the verifier. When neither guarded event of §3 occurs, the circuit computes exactly Plonky3's verifier; we test this op for op against Plonky3's logged transcript and on real proofs. When a guarded event occurs, the circuit outputs 0. Hence an output of 1 implies that Plonky3's verifier accepts, and an honest proof is rejected only if a guarded event occurs, with probability at most $\varepsilon_{\text{guard}} \approx n \cdot 2^{-127}$ for n guarded coordinates.

Soundness of the proof system. We use Plonky3's own security assessment of each configuration as the input to our classical accounting. The proximity terms are in the Johnson list-decoding regime (Table 2). The quantum conversion in §5.1 additionally assumes round-by-round soundness for the sumcheck-based layers. At 2^{18} rows the binding term is the WHIR opening at 104.3 bits (110 bits per term, rate 1/32, folding factor 4, proof of work up to 32 bits). The other terms are constraint batching 109.0, zerocheck 114.5 and its sumcheck 113.5, bit ring switch 114.0, column batching 115.2, and commitment and transcript collisions 128 (Blake3). We write $\varepsilon_{\text{STARK}}(t)$ for the composed error against an adversary making t hash queries. Plonky3's report states it as bits of security, 104.3 at 2^{18} rows, meaning that $\varepsilon_{\text{STARK}}(t) \leq t \cdot 2^{-104.3}$ for a classical adversary; §5.1 gives the quantum bound.

Authenticity of the garbling. Our scheme is privacy-free garbling [FNO15] with free-XOR [KS08] and one ciphertext per non-free gate [ZRE15], with H = Blake3 and a fresh secret Δ per garbling. Free-XOR is secure when H is circular correlation robust [CKKZ12]; we assume this of Blake3, as is standard for hash-based garbling. In the BitVM setting the garbled circuit is published at setup and the proof is chosen afterwards, so authenticity must hold for inputs chosen adaptively, after the garbled circuit is seen [BHR12a]. Let ε_{GC} be the advantage of an evaluator that sees the garbled circuit, then chooses an input x, receives its labels, and outputs the label of the value $\neg C(x)$ on the output wire. We assume that our Blake3-based free-XOR garbling has adaptive authenticity in this sense, with Blake3 modelled as a random oracle; Bellare, Hoang and Rogaway give constructions proven adaptively secure [BHR12a].

trace rows	2^5	2^8	2^{12}	2^{16}	2^{18}	2^{20}
composed soundness (bits)	106.1	105.2	104.7	104.3	104.3	104.0

Table 2: Soundness stays at about 104 bits at every measured size, 103.95 at 2^{20} rows. The values are Plonky3's composed soundness for each configuration.

The following circuit-level statement is conditional on the Blake3 circular-correlation-robustness and adaptive-authenticity assumptions stated above. It does not cover cut-and-choose, input soldering or Bitcoin transaction enforcement.

Theorem 1. *Let C be a correctly garbled verifier of a fixed configuration, and x the encoding of a proof π for a statement.*

(i) Soundness. *If the statement is false, an evaluator holding the labels of x obtains the output label of value 0, except with probability at most $\varepsilon_{\mathrm{STARK}}(t)$, where t bounds the hash queries of the party that chose π.*

(ii) Authenticity. *If π is honestly generated, an evaluator holding the labels of x obtains the output label of value 0 with probability at most $\varepsilon_{\mathrm{guard}} + \varepsilon_{\mathrm{GC}}$.*

Proof sketch. (i) If $C(x) = 1$, Plonky3's verifier accepts π by the previous paragraphs, which for a false statement happens with probability at most $\varepsilon_{\mathrm{STARK}}(t)$. Otherwise $C(x) = 0$, and correctness of the garbling yields the label of value 0. (ii) For an honest proof $C(x) = 1$ unless a guarded event occurs, and the label of value 0 is then the label of $\neg C(x)$, which the evaluator obtains with probability at most $\varepsilon_{\mathrm{GC}}$. □

The theorem applies only to the relation represented by C. In our experiments that relation is Keccak-f trace validity without external public values; it does not prove authorization of a bridge withdrawal.

Section 6 uses (i) to protect challengers against a false claim and (ii) to protect an honest operator.

5.1 Post-quantum analysis

The verifier construction avoids number-theoretic assumptions: the proof system, commitments, transcript and garbling use Blake3. This statement does not cover Bitcoin transaction authorization or every proposed input-publication mechanism. The estimates below are conditional on the stated random-oracle, round-by-round-soundness and adaptive-authenticity assumptions; they are not an end-to-end post-quantum security proof. Pairing-based verifiers (Groth16, Pari and their designated-verifier variants), the witness-encryption schemes built on them (§8) and Bitcoin's own Schnorr signatures fall to Shor's algorithm [Sho94]. Our verifier does not. Avoiding discrete-logarithm and pairing assumptions is necessary for quantum resistance here, but a hash-based verifier is not sufficient by itself: a quantum adversary still speeds up generic preimage and collision search [Gro96, BHT98]. Table 3 gives component-level estimates.

Fiat–Shamir against quantum adversaries. In the quantum random-oracle model, Chiesa, Manohar and Spooner [CMS19] show that applying Fiat–Shamir with a hash to a public-coin interactive oracle proof with round-by-round soundness error $\varepsilon_{\mathrm{rbr}}$ bounds the success of a quantum adversary that makes t hash queries by $O(t^2\varepsilon_{\mathrm{rbr}} + t^3/2^{\lambda})$, where $\lambda = 256$ is the digest length. A classical adversary is bounded by $O(t\,\varepsilon_{\mathrm{rbr}})$.

WHIR has round-by-round soundness [ACFY25, §5]. For the sumcheck-based layers (zerocheck, column batching, ring switch) we assume it, as is standard for sumcheck, with each of Plonky3's terms as the round error. A round with error 2^{-a} therefore offers about $a/2$ bits against a quantum adversary, and Grover search [Gro96] attains this generically when the bad challenges form an unstructured set. Proof-of-work grinding is generic search too: b bits of grinding cost 2^b hashes classically and about $2^{b/2}$ quantumly.

The proof system at our parameters. At 2^{18} rows each WHIR term targets 110 bits, of which grinding supplies 26 to 32 bits (per round: 33 queries with 30 and 28 bits of grinding, 20 with 32 and 30, 15 with 29 and 32, and 12 terminal queries with 27). Both parts halve under quantum search, so each WHIR term offers about 55 bits, and the composed 104.3 bits become about 52. The algebraic terms (zerocheck 114.5, its sumcheck 113.5, ring switch 114.0, column batching 115.2, constraint batching 109.0) are Schwartz–Zippel errors over the 2^{128}-element field and offer 54 to 58 bits. They are capped by the field: no number of queries raises them. Collisions in the 256-bit Blake3 digests cost about 2^{85} quantum queries [BHT98], and preimages 2^{128}.

The ceiling of the current configuration space. Raising Plonky3's per-term target does not raise the quantum level. We searched the target that the proven-list-decoding profile accepts at 2^{18} rows. Up to 134 bits per term it keeps the queries at $33 + 20 + 15 + 12$ and raises the grinding to 51–56 bits, its ceiling; at 200 bits it refuses, needing 103 bits of grinding. Plonky3's security report fully assesses targets up to 117 bits per term, where the composed soundness is 108.6 bits, bound by constraint batching at 109.0, a field-size term no WHIR parameter moves. Grinding and the field-bound terms both halve under quantum search, so no configuration of the current proof system exceeds about 54 quantum bits. Reaching a quantum target is a change to the proof system, not to its parameters.

The garbling. Authenticity rests on the secrecy of Δ, and Δ can be tested against a single gate. An evaluator holding the labels a, b of an AND gate with known values x, y and its ciphertext ct checks a candidate Δ' by recomputing ct from $a, a \oplus \Delta', b$ and Δ'. A wrong candidate passes with probability 2^{-128}. Grover search therefore recovers Δ with about 2^{64} quantum evaluations of Blake3, after which every label's complement is known and the output label of the other value is free. With 128-bit labels the garbling offers about 64 bits against a quantum adversary. The same holds for a hashlock on a 128-bit output label: a preimage search over 2^{128} labels costs about 2^{64}. Grover search is highly sequential, and the adversary has only the time between setup and the end of the dispute window, but we do not rely on this.

The surrounding protocol. The protocol of §6 publishes input bits under one-time keys. Lamport and Antichain Winternitz keys rest on `HASH160` preimages. The familiar 2^{64} quantum cost for our 16-byte preimages is a single-target estimate. One Lamport key set exposes $T = 1{,}041{,}024 \approx 2^{20}$ alternative preimages, so generic multi-target search costs about $2^{128}/T \approx 2^{108}$ classically and $\sqrt{2^{128}/T} \approx 2^{54}$ quantumly. Seven independent sets lower these figures to about $2^{105.2}$ and $2^{52.6}$. Hash-chain alternatives have the same multi-target issue; Table 5 gives the widths that lift the selection itself above this bound. Even 20-byte preimages under `HASH160` are capped near 70 quantum bits for one set, rather than 80, once the target count is included. The surrounding protocol is not post-quantum as instantiated. Schnorr adaptor signatures, the committee's pre-signed transactions and Taproot key-path spends fall to Shor's algorithm. Hash-based input keys avoid the adaptor-signature dependency, but their one-key-set cost assumes the unimplemented soldering in

component	attack	classical	quantum estimate	to reach λ_Q quantum bits
WHIR opening	Fiat–Shamir (FS) search, grinding	104.3 (max. 111.3)	≈ 52	$\approx 2\lambda_Q$ per term: $\approx 2.1\times$ queries at $\lambda_Q = 100$
zerocheck, sumchecks, ring switch	FS search	109–115	≈ 54–58	a field of $\gtrsim 2^{2\lambda_Q+14}$ elements: $\mathbb{F}_{2^{256}}$
Merkle and transcript hash	collision, preimage	128	≈ 85	$\gtrsim 3\lambda_Q$-bit digest and wider internal state; unmeasured
garbling (Δ, output hashlock)	Grover on 128-bit labels	128	≈ 64	$2\lambda_Q$-bit labels: 256 bits
hash-based input keys	multi-target preimage, $T \approx 2^{20}$ ($7T$ for seven sets)	≈ 108 (105.2)	≈ 54 (52.6)	at least 220 (223) bits of preimage entropy under a 256-bit hash
cut-and-choose $(181, 7)$	bad kept set	40	40	increase (N, M) for the target error
adaptor keys, pre-signed graph	Shor	–	broken	post-quantum transaction authorization

Table 3: The measured proof system has a bottleneck near 52 quantum bits under the stated assumptions; this is not the composed protocol level. The reported cut-and-choose parameters provide about 40 statistical bits, and the transaction authorization is broken by Shor. Reaching 100 quantum bits also requires more queries, a 256-bit field, 256-bit labels, a hash with at least a 300-bit digest and sufficiently wide internal state, a 256-bit input hash with 28-byte preimages, stronger cut-and-choose and post-quantum transaction authorization.

§6; the seven-key-set projection is 37–54 times the adaptor payload. Bitcoin has no opcode that natively verifies a post-quantum signature. QSB gives a non-standard hash-based authorization path under existing consensus rules [Lev26, Sta26], but whether it can replace the authorization of a pre-signed, timelocked dispute graph is outside our scope. We therefore treat graph authorization as an explicit open assumption rather than a solved component.

Parameters for a quantum target. There is no single rule that doubles every classical parameter. Fiat–Shamir and preimage-search terms need about $2\lambda_Q$ classical bits, generic collision search on an n-bit digest costs about $2^{n/3}$ quantum queries, and an information-theoretic cut-and-choose error must itself be at most $2^{-\lambda_Q}$. Seven changes follow.

- The WHIR terms need about $2\lambda_Q$ bits each. Since grinding also halves, the query count must carry most of it: from about 80 information-theoretic bits per term today to about 168 at $\lambda_Q = 100$, about $2.1\times$ the queries. Plonky3's profile raises grinding instead, so this needs a query-heavy profile.
- The algebraic terms need challenges from a field of about $2^{2\lambda_Q+14}$ elements, which means one more tower level, $\mathbb{F}_{2^{256}}$, at $3^8 = 6{,}561$ AND gates per multiplication, three times today's. Repeating each algebraic check with independent challenges is an alternative at a similar cost.
- Labels of $2\lambda_Q$ bits: 256-bit labels double the ciphertext size to 32 bytes per non-free gate. The gate count is unchanged, because labels feed the garbling hash, not the circuit.
- Merkle commitments and the transcript need a hash that withstands 2^{λ_Q} quantum collision search. Generic collision search therefore calls for at least a $3\lambda_Q$-bit digest and a sufficiently

wide internal state. Blake3-256 caps this component near 85 quantum bits; requesting a longer XOF output does not widen its 256-bit chaining value. A replacement hash circuit and its cost remain to be designed and measured.

- For T possible input-authentication targets, 100 quantum bits require at least $200 + \lceil \log_2 T \rceil$ bits of preimage entropy under a hash wider than 160 bits: about 220 bits for one key set and 223 for seven. Byte-aligned 28-byte preimages under `SHA256` meet this generic target, but that construction and its larger witness cost remain to be designed and measured.
- Cut-and-choose needs parameters satisfying $1/\binom{N}{M} \leq 2^{-\lambda_Q}$, together with an ungrindable uniform selection of the opened instances. The setup time and kept-instance storage must be recomputed for those parameters.
- The pre-signed graph needs post-quantum transaction authorization. Bitcoin currently has no general post-quantum signature opcode. QSB gives a non-standard hash-based spend under existing consensus rules, but adapting it to the committee-authorized, timelocked graph is unstudied here.

Applying only the first three factors to the measured 2^{18} verifier profile gives a partial lower-bound estimate, not a 100-bit configuration measurement. Merkle paths and leaf hashes scale with the query count (2.1×), and leaves also with the element size (2×), to about 45M gates. The transcripts absorb twice the bytes, about 24M. Multiplications triple, and those that depend on the queries also scale with them, to about 213M. The result is about 280M non-free gates, 3× today's, and about 9 GB garbled at 32 bytes per gate. It leaves Blake3-256 and its 85-bit quantum collision ceiling unchanged and excludes the replacement hash circuit, stronger cut-and-choose, input authentication and graph authorization. It is therefore not an estimate of a 100-bit end-to-end protocol.

6 Bitcoin integration and cost boundaries

This section sketches how the circuit output could be consumed by a BitVM3-style dispute and isolates the accounted components of publishing its input. We use *total on-chain footprint* to mean the sum of the serialized virtual sizes of every transaction in the dispute graph, including ordinary transaction signatures, authentication signatures or preimages, witness framing, revealed tapscripts, control blocks, transaction inputs and outputs, anchors, timeouts and fee-management transactions. We have not constructed, serialized or policy-tested that graph, nor implemented hash-based soldering across the kept garblings. We have constructed and policy-tested bounded funding–reveal publication slices for all three input schemes, and a signed join for Lamport, but not the challenge, timeout, Disprove, anchor or fee-management branches around it. We therefore do not report a total on-chain cost. The figures below are measurements and conditional projections of the input-publication component, not evidence that the complete construction is enforceable on today's network. We take BitVM3's roles [WAA+26, KTA+26]: the operator garbles, and challengers evaluate. A bridge is the motivating application, not an implemented statement. Table 4 separates executed components from open integration work.

Setup. The operator garbles N instances of C from independent seeds. Before any instance is opened, it commits for every instance j the hash of its garbled tables, $h_j = \mathrm{SHA256}(\ell_0^j)$, the hash of the label of output value 0 (reject), and the instance's input-key material: the hashes of its one-time keys and its label translations. A challenger then chooses $N - M$ instances to open, re-garbles each opened instance from its seed and checks every commitment of that instance; only then is

component	status	evidence or missing work
Keccak-f/WHIR verifier circuit	implemented, measured	real proofs, transcript equivalence and gate-level evaluation
streaming garbler and evaluator	implemented, measured	gate counts, runtime and live-wire memory
one-copy Script components	executed	Lamport publication, label translation and reject-label hashlock
transaction-bound authentication fixtures	serialized, Core-tested	complete signed adaptor, Antichain and Lamport reveal-transaction fixtures with real sighashes, negative tests and strict-policy replay after funding confirmation
bridge public-input binding	not implemented	the measured AIR has no public values
hash keys across M garblings	not implemented	zero-knowledge soldering or an independently proved fallback is required
transaction-bound Lamport publication	serialized, Core-tested	1,044 funded leaves, 174 signed reveals and a signed join; confirmations must separate the stages
complete dispute graph	not implemented	completion semantics, challenge/Disprove and timeout branches, anchors and fee management are required

Table 4: The implementation validates the verifier and one-copy Script components, not an end-to-end Bitcoin protocol.

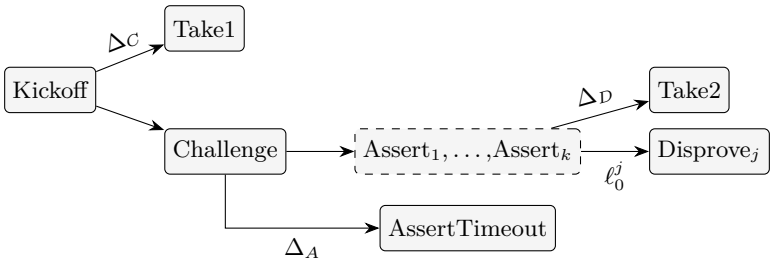


Figure 2: Logical outcomes of the proposed dispute. The dashed Assert$_1$, . . . ,Assert$_k$ node denotes an unspecified publication subgraph. The figure does not define its UTXO dependencies, completion anchor, timeout enforcement or fee management, all of which remain to be constructed and policy-tested.

the transaction graph signed. The streaming garbler re-garbles in one garbling pass per instance, reusing the plan that all instances share (§4). The challenger stores the M kept instances, 1.5 GB each at 2^{18} rows. The intended protocol fixes one-time keys for the input bits of C. Publishing bit b of input i must give a challenger only the label of value b on wire i in every kept instance. Mosaic provides this correlation for its adaptor-signature path [KTA+26]; the corresponding hash-based soldering is open here. The intended transactions would be pre-signed at setup by a committee, as in BitVM2 [LAA+25], whose enforcement needs one honest signer.

Figure 2 summarizes the intended logical outcomes for one claim. The Assert node is a placeholder for an unconstructed transaction subgraph.

Required transaction semantics. The intended branches are:

- *Kickoff*: the operator claims reimbursement for a withdrawal it has paid. If no challenge follows within Δ_C, *Take1* pays it. This is the happy path.

- *Challenge*: any challenger may spend Kickoff's challenge output, paying the operator a bond that covers the cost of Assert, as BitVM2's crowdfunded challenge does [LAA+25].
- $Assert_1, \ldots, Assert_k$: the operator must publish every authenticated circuit-input bit. Because the payload exceeds one standard transaction, a concrete construction needs a subgraph whose terminal output is spendable only after every chunk has been published. We do not specify that fanout, chaining or completion mechanism here.
- *AssertTimeout*: if the operator fails to create the completion output before Δ_A, a timeout branch should transfer its stake. The enforceable condition for "all chunks published" depends on the missing subgraph and is not specified by Figure 2.
- $Disprove_j$: after such a completion output exists, a challenger holding ℓ_0^j would open the corresponding hashlock before Δ_D; otherwise Take2 would pay the operator. The required transaction dependencies and relative timelocks remain unimplemented.

All three authenticated rows below now serialize complete signed reveal transactions and execute their selected leaves with real BIP341 sighash contexts. Each fixture uses BIP341's unknown-discrete-log NUMS internal key, so there is no known key-path secret that bypasses the measured script leaf. Their output shapes still differ, but all three sets have been replayed through Bitcoin Core's strict standard-policy mempool after funding confirmation; only the Lamport fixture includes a signed join. Additional graph inputs and outputs, non-authentication scripts and control blocks, graph-authorization signatures, anchors, timeouts, dependencies and fee-management overhead are not uniformly included. Current Bitcoin Core policy limits a standard transaction to 400,000 weight units and an unconfirmed cluster to 101 virtual kilobytes [Bit26]; the complete graph must be tested against the release policy targeted for deployment.

The evaluated Keccak-f circuit has no public values and therefore does not instantiate the bridge statement. A bridge deployment must bind the withdrawal, rollup-state commitment and other public claim data into both the AIR relation and Fiat–Shamir transcript. We have not implemented or measured that extension; its circuit and input costs are outside the reported figures.

Intended dispute. In a completed instantiation, a challenger would read the published bits, obtain the corresponding input labels for every kept instance, and evaluate each instance. Evaluation takes one Blake3 call per non-free gate and runs off-chain: 17 s at 2^8 rows (§4). If the proof is invalid, a correctly garbled instance yields ℓ_0^j, which is intended to open Disprove_j. The circuit size affects off-chain evaluation, while the proof size controls the Assert payload. This flow is conditional on the missing soldering and transaction subgraph.

Security. Let ε_{cc} bound the probability that every kept instance is malformed while every opened one is correct; for choosing $N - M$ of N uniformly this is $1/\binom{N}{M}$, which is 2^{-40} at $(N, M) = (181, 7)$ [GKS+26]. Let ε_{sol} bound a failure of the soldering, and let ε_{sig} bound the aggregate probability of forging any one-time or transaction-authorization signature across the entire dispute. It is not a per-key probability; a reduction from a per-target bound must include the number of available targets. At these parameters, cut-and-choose alone limits the intended malicious-garbler security to about 40 bits, irrespective of the STARK verifier's 104-bit classical soundness. Subject to an enforced transaction graph, timely confirmation and an instantiated soldering bound, Theorem 1 supports the following intended properties.

- *A false claim is disproved.* If the operator's statement is false, an honest challenger obtains some ℓ_0^j except with probability $\varepsilon_{\text{STARK}}(t) + \varepsilon_{\text{cc}} + \varepsilon_{\text{sol}}$: a correctly garbled kept instance

exists, its labels are those of the published proof, and Theorem 1(i) applies. An operator that publishes no proof loses its stake to AssertTimeout.

- *An honest operator keeps its stake.* If the operator's proof is honest, no challenger obtains any ℓ_0^j except with probability $M(\varepsilon_{\text{guard}} + \varepsilon_{\text{GC}}) + \varepsilon_{\text{sol}} + \varepsilon_{\text{sig}}$, by Theorem 1(ii) and a union bound over the kept instances.

This is conditional accounting, not a protocol theorem in a ledger model: ε_{sol} has no instantiated bound, and the transaction graph and ledger-timing assumptions have not been implemented or proved. BitVM3 [WAA+26] and BABE [GKS+26] analyze their own instantiations.

Label selection. Every input mechanism below is an instance of one two-party task, which we state once so that the three baselines and the hash-based alternative can be compared on the same terms. The *garbler* G (the operator, who also proves) holds, for every input wire w of the kept garblings, the two labels ℓ_w^0, ℓ_w^1. The *evaluator* E (a challenger) will run the garbled verifier on the proof π. Publishing π on-chain must therefore (i) *deliver* to E the label $\ell_w^{\pi_w}$ of every input wire, (ii) be *exclusive*: no party learns $\ell_w^{1-\pi_w}$ for any w, since a single complementary label breaks authenticity (§5), (iii) be *binding*: G can publish at most one value per wire, and the publication is at the same time the authorization of the spending transaction, and (iv) be *public*: any observer can recover π and check that the published object is well formed. A malformed publication does not satisfy the Assert script and cannot be spent, and a missing one forfeits the operator's stake through AssertTimeout; Disprove is reserved for the reject label. Lamport achieves (i)–(iv) with one preimage per bit, and its cost is the per-bit script. Figure 3 is the adaptor construction of BitVM3 and Mosaic [WAA+26, KTA+26], which achieves them with one signature per k-bit digit and no per-bit script. Figure 4 replaces the signature by a pair of hash-chain openings and is the construction behind the Antichain row and the hash-based projections of §5.1.

The two protocols publish the same amount of selection material per digit: one 64-byte completion for Figure 3, two n-byte nodes for Figure 4. With the adaptor at $k = 8$ and the hash-based fixture at $k = 4$, $n = 16$, both are 8 bytes of witness per bit, and indeed the 130,128 adaptor completions and 16,266 64-bit hash records of our fixtures are both exactly 8,458,320 bytes. They differ in where the remaining cost sits. The adaptor's selection is verified by one `CHECKSIG` per digit, 3 bytes of script, and its binding is the same signature; its setup holds 2^k secrets and ciphertext bundles per digit, and its exclusivity rests on discrete logarithms, which Shor's algorithm removes (§5.1). The hash-based selection has a two-node setup per digit and hash-only exclusivity, but under current consensus every hash step of the verification is 2 bytes of script, plus a separate transaction signature. That script is what makes the Antichain row 5.3 times the adaptor row in Table 9; the digit-to-label translation table is the same object in both protocols and is not on-chain.

A post-quantum instantiation. Figure 4 becomes a post-quantum replacement for the adaptor under three changes, none of which is available on Bitcoin today. First, the forward hashing moves out of Script into a native batch verifier: one leaf per transaction carries the batch root ρ, the first digit index and the count, and an opcode re-derives every endpoint from the records on the stack. Second, the transaction-binding signature becomes hash-based; the draft P2MR output type and its 0xC2 leaf version[1] would provide a 548-byte stateful hash-based signature per transaction and 6,000-byte stack items. Third, the node width n must match the quantum target: forging a different digit from a published opening is a second-preimage search over the roughly 2^{22} chain

[1] BIP360, `https://github.com/bitcoin/bips/blob/master/bip-0360.mediawiki`, and the SHRINCS proposal for a stateful hash-based signature in that leaf; both are drafts and may change.

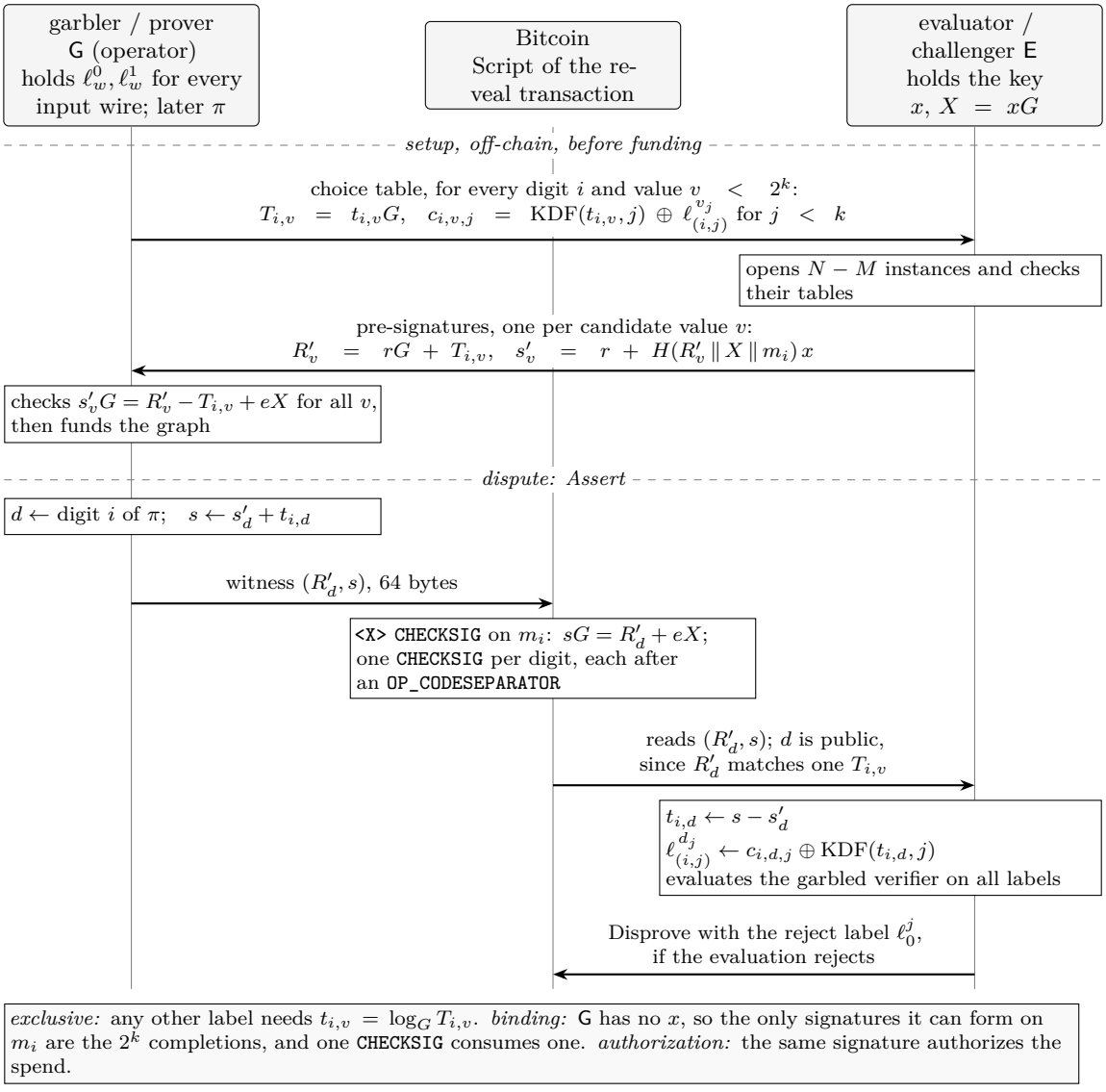


Figure 3: Schnorr adaptor label selection for one k-bit digit position i, as in BitVM3 and Mosaic [WAA+26, KTA+26]. m_i is the BIP341 message seen by the i-th CHECKSIG; wire (i, j) is bit j of the digit. Everything above the dispute line is off-chain and repeated for every digit and every candidate value.

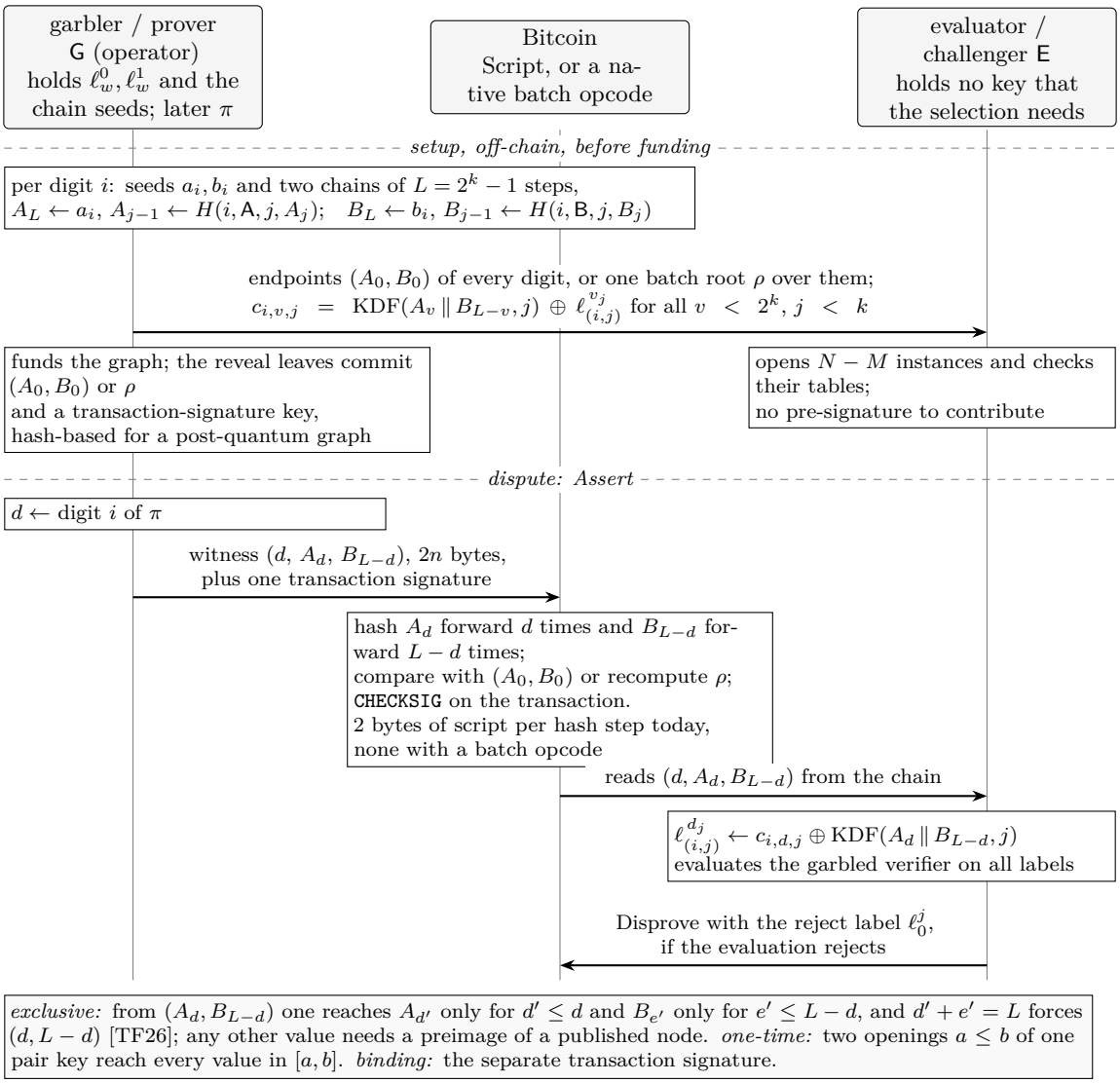


Figure 4: Hash-based label selection for one k-bit digit position i with n-byte chain nodes. The selection material is two nodes instead of one signature, the evaluator contributes nothing at setup, and the transaction binding that the adaptor gets from the same signature must be added separately. Verification is a hash ladder, in Script today or in a batch opcode under the soft fork of Table 5.

selection	consensus	n	k	vB/bit	component	selection security, quantum
Schnorr adaptor (Fig. 3)	current	–	8	2.14	2.23 MvB	broken (Shor)
Antichain $(2, 16)$ in Script (Fig. 4)	current	16	4	11.38	11.88 MvB	≈ 53 bits
batched selector, one record per item	P2MR, 0xC2, batch opcode	16	4	2.05	2.13 MvB	≈ 53 bits
batched selector, 11-record items	same	16	4	2.04	2.12 MvB	≈ 53 bits
batched selector, 5-record items	same	24	4	3.05	3.17 MvB	≈ 85 bits
batched selector, projected	same	32	8	≈ 2.05	≈ 2.13 MvB	≈ 117 bits

Table 5: Hash-based label selection against the adaptor baseline for the 1,041,024-bit input, funding plus reveals. The first two rows are the Core-tested fixtures of Table 9. The batched-selector rows are exact rust-bitcoin serializations of a construction that requires the draft P2MR output, a hash-based transaction signature and a hypothetical batch verification opcode; the last row is a model projection. The security column is the multi-target second-preimage cost of the chain nodes alone; the composed protocol remains bound by the proof system (§5.1).

nodes that the whole input makes public, so $n = 16$ gives about 53 quantum bits, $n = 24$ about 85 and $n = 32$ about 117. We serialized this construction exactly with rust-bitcoin for $n = 16$ and $n = 24$ at $k = 4$, including the P2MR outputs, 0xC2 leaves, placeholder signatures, depth-one control blocks and canonical multi-record stack items, and tested the pair code and the batch parser against tampering, reordering, cross-batch splicing and padding; nothing in it is executable by Bitcoin Core. Table 5 gives the results together with a projection whose model reproduces both serialized fixtures within 0.03%. The operative row is $n = 32$, $k = 8$: full-width nodes, about 117 quantum bits for the selection, and the same 2.13 MvB as the Schnorr adaptor, at the price of up to 510 hash steps per byte inside the opcode, about 6.6×10^7 compressions for the input. Its setup is two seeds per digit rather than the adaptor's 2^k secrets, so widening k does not grow the choice table; only the translation table grows, to $2^k \cdot k \cdot 16$ bytes per digit, 4.3 GB per instance at $k = 8$. The delivery and soldering of that table across the M kept garblings is the same open item as for every other row.

Publishing the input. We evaluate three baseline one-time schemes for carrying the input bits, and they trade on-chain size against assumptions.

- *Lamport.* Each bit has two hashed 16-byte preimages, and the revealed preimage is the one-time-signature witness. The leaf script per bit,

  `OP_HASH160 OP_DUP` h_0 `OP_EQUAL OP_SWAP` h_1 `OP_EQUAL OP_BOOLOR OP_VERIFY`

  is 49 bytes, and the witness 17 bytes. We executed it with 998 bits per input, and it rejected wrong preimages. The transaction-bound fixture prefixes each leaf with a 32-byte authorization key and `OP_CHECKSIGVERIFY`, and supplies a 64-byte `SIGHASH_DEFAULT` signature. Its initial stack has 999 items and the measured transient peak is exactly the 1,000-item Tapscript limit [WNT20]. Complete one-output reveal transactions give 16.59 vB per input bit. The reported row uses a one-leaf tree with a 33-byte control block; timeout siblings and their longer control paths belong to the missing complete graph.

- *Antichain Winternitz.* Plain Winternitz chains are unsafe here: from a published chain value the evaluator hashes forward to the labels of other digit values [TF26]. Antichain Winternitz closes this and costs a cited marginal 11.31 vB per bit at parameters $(2, 16)$, counting the hash-chain opening witness and authentication script but cancelling fixed transaction overhead. Our transaction-bound fixture pins the upstream implementation, prepends a 64-byte Schnorr authorization, and includes two P2TR outputs, input shells, leaf scripts and depth-zero controls. With at most 332 four-bit digits per input, 784 inputs in 131 transactions cost 11,848,875 vB in the all-nonzero worst case, or 11.38 vB per input bit. Zero-valued ScriptNum coordinates only reduce this figure. The source reports 0.256 kB of off-chain table per bit. Its larger parameters save more on-chain but need up to 43.8 kB per bit off-chain [TF26], 46 GB for our input.

- *Schnorr adaptor signatures* over 8-bit digits, as in BitVM3 and Mosaic. BitVM3 reports 9,707 weight units (WU) for a 1,024-bit Assert transaction with one input and two P2TR outputs [WAA+26], or 2.37 vB per bit at that size. Its printed witness formula instead sums to 9,451 WU once the SegWit marker and flag are included, while the merged 64-byte-`SIGHASH_DEFAULT` implementation implies 9,323 WU for that shell. Our executable fixture reproduces 9,323 WU, uses the BIP342 opcode index rather than a byte offset for every code separator, and reaches the exact 998-signature stack capacity. Core 31.1 validates all completed signatures. Integer packing of 130,128 eight-bit digits uses 133 inputs in 23 two-output transactions and totals 2,222,213 vB, or 2.13 vB per input bit. Using the minimum 131 inputs instead needs 26 transactions and is 184 vB larger. Mosaic's source construction solders all kept instances to one set of signatures by polynomial label correlation, with an on-chain footprint independent of the number of instances [KTA+26]. This repository tests only the one-key-set transaction fixture; it does not implement or test Mosaic's polynomial correlation or setup verification. The signatures rest on discrete logarithms, and an evaluator that holds the signing key can spend without completing the adaptor [TF26].

Packing more bits into each adaptor completion reduces the serialized transaction shape. Our Core-tested 16-bit fixture uses 2,960 vB of funding and 1,111,128 vB across 12 reveals, or 1,114,088 vB in 13 transactions. Exact serializations of 10-, 11- and 12-bit digits, still with placeholder signatures and not replayed in Core, give 1,782,498, 1,620,499 and 1,485,410 vB including funding. The price is off-chain: a table of every digit's 2^b choices at 65 bytes each is 2.165 GB per key set at 8 bits, 6.929, 12.598 and 23.097 GB at 10, 11 and 12 bits, and 277.162 GB at 16. This is an encoding sensitivity result, not yet a binary-label authentication mechanism: the complete 2^{16} choice table and exclusive digit-to-binary-label delivery remain unimplemented. We therefore retain the 8-bit construction as the authenticated baseline in Table 9. Our labels are free-XOR correlated, so a hash-based preimage s cannot itself be a label. A key-derivation function (KDF) maps it to one by a translation ciphertext $\mathrm{KDF}(s) \oplus w_b$, two per input bit and instance: 32 bytes per bit, or 33 MB per instance at 2^{18} rows. Using one on-chain key set for all M kept instances requires a proof that the translations of both opened and kept instances encode the same bit choices. BABE's zero-knowledge soldering [GKS+26] and Glock's verifiable secret sharing [Eag25] provide related mechanisms at smaller input sizes, but we have neither constructed nor costed one for a million input bits. Consequently the one-key-set hash rows in Table 9 are conditional on an unimplemented mechanism, and $\varepsilon_{\mathrm{sol}}$ is presently unbounded.

A conservative component projection authenticates each input independently for every kept instance, multiplying the accounted authentication component by M. This is not itself an implemented or proved protocol because the same bit choice must still be enforced across all key sets.

At $M = 7$, simple multiplication projects the accounted Lamport component to 120.86 MvB and the all-nonzero Antichain component to 82.94 MvB, before transaction-graph overhead.

Executed components. We ran the dispute components on a small circuit, with its on-chain steps executed as Bitcoin Script by `bitcoin-scriptexec`. The statement is knowledge of a 64-byte x with Blake3(x) = h: one Blake3 compression, 512 input bits and 10,536 non-free gates. The operator garbles it at setup, before choosing x, and publishes Lamport hashes, translation ciphertexts, the labels of the two constant wires and SHA256 of the reject label. Its Assert publication runs through a 25 KB Lamport script under the 1,000-item stack limit. The challenger derives the input labels from the published preimages and evaluates the stored garbling without Δ or any label it was not handed. For a true x it obtains the accept label, which the Disprove hashlock script rejects; for x with one bit changed it obtains the reject label, which opens it. Assert rejects a preimage outside the committed keys. We then ran the same dispute on the Keccak-f STARK verifier at 2^5 rows. Garbled with every input at 0, it reads 586,240 input bits, which the operator publishes through 588 Assert scripts of at most 998 bits each, 28.7 MB of script in all, each executed under the stack limit. The challenger's evaluation of the stored garbling from the published labels gives the accept label on the real proof and the reject label, which opens Disprove, on the proof with one opened value changed. Signatures, timelocks and cut-and-choose are not part of either run. These runs validate individual input-publication, label-translation, evaluation and hashlock components for one garbling. They do not validate Figure 2, multi-instance soldering, transaction policy or ledger-level security.

7 Evaluation

The experiments answer five questions: how the garbled verifier scales with the trace (Table 6, Figure 5), where its gates go (Table 7), how much each design choice contributes (the ablation), how the circuit-input authentication component projects and how far it shrinks (Tables 9 and 10), and how it compares with pairing-based designs (Table 12).

Setup. We proved Keccak-f permutations with Plonky3's 1,625-column characteristic-2 Keccak AIR (booleanity is not constrained, since the commitment is to bits). A byte-logging challenger recorded each run, our reference replayed Plonky3's verifier byte for byte, and the measured verifier circuit was then streamed through the garbler. The parameters were a rate of 1/32, a folding factor of 4, and a target of 110 bits per WHIR term. Garbling was single-threaded. Every run up to 2^{18} rows was allocated 32 GB of RAM and eight cores on an Intel Xeon Platinum 8375C at 2.9 GHz; the 2^{20} run, whose proof took longer to generate, ran on an AMD EPYC 9355 limited to 128 GB and 32 cores.

Artifact availability. The source repository includes the locked dependencies and benchmark commands; the commands and memory caps are listed in `whir-gc/README.md`. The raw measurement logs, with a SHA-256 manifest and a map from tables and figures to logs, are archived separately and available on request. The experimental Plonky3 and Ziren changes behind Table 11 are Ziren commit `80af2779` on `feat/binary-whir-blake3` and Plonky3 commit `be1c6f92` on ProjectZKM's `zkm/binary-whir`; `patches/README.md` lists them and the commands that rebuild the tapes. `Cargo.lock` pins Plonky3 at revision `ca4da2daa2e0dfdd31419af5c870813083398955` and bitvm-gc at revision `10083d4a63eb08a344dc40ee84bc11394b218a89`. Gate, wire and input counts

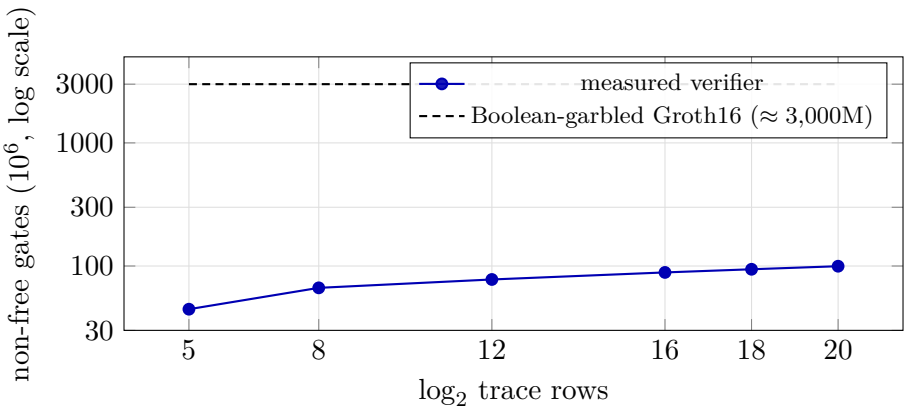


Figure 5: On a logarithmic vertical axis, the measured verifier grows from 44.6M non-free gates at 2^5 rows to 99.6M at 2^{20}, a 2.2× increase for a 32,768× larger trace. The dashed line shows the approximately 3,000M-gate Boolean-garbled Groth16 comparator.

are deterministic; wall-clock times are single-run engineering measurements on the machines above unless a range is stated.

The circuit's size depends on the AIR: the transcript, the column combination and the constraint evaluation grow with the number of columns and constraints. All sizes below are therefore for this 1,625-column AIR, which is a demanding case. For a fixed verifying key and public-input arity, a Groth16 verifier's circuit does not grow with the proved trace length.

Scaling. Table 6 and Figure 5 give the measured verifier configuration from 2^5 to 2^{20} rows. Every circuit accepted its proof, and at 2^5 and 2^8 we also checked that it rejected the proof with one opened value changed. Growing the trace 32,768-fold grew the circuit 2.2-fold, because the verifier depends on the trace length only through the logarithmic number of variables.

Garbling efficiency. To compare garbled verifiers across trace sizes and AIRs, we define the *garbling efficiency* of a configuration as

$$\eta = \frac{|\mathsf{GC}|}{R \cdot W} \quad \text{bytes per committed cell,}$$

where $|\mathsf{GC}|$ is the garbled circuit's size in bytes, 16 per non-free gate, R the number of trace rows and W the number of committed columns. It is a cost per unit of verified computation, so smaller is better. For our Keccak-f AIR, $W = 1{,}625$ and each cell is one bit. Because $|\mathsf{GC}|$ grows with $\log R$ while $R \cdot W$ grows with R, η falls almost in inverse proportion to the trace: from 13,715 bytes per cell at 2^5 rows to 3.53 at 2^{18} and 0.94 at 2^{20} (Table 6). At 2^{20} rows this is less than one garbled byte per committed bit: 7.5 garbled bits per bit, or 38 kB per Keccak-f permutation. A verifier whose size is fixed with respect to trace length, such as Groth16 for a fixed verifying key and public-input arity, has η inversely proportional to the computation its proof covers as well; comparing η across proof systems therefore needs the same computation and cells of the same kind, bits here against field elements in a prime-field AIR.

Where the gates go. Table 7 breaks the 2^{18} circuit down by the verifier phase that emits each gate. The multi-STARK layers account for 70.8% and WHIR for 29.2%. Three of the largest items,

trace	vars	non-free gates	garbled	η (B/cell)	garbling	garbler memory	input bits
2^5 rows (1 Keccak-f)	9	44,573,121	713 MB	13,715	35 s	0.32 GiB	586,240
2^8 rows (10)	12	66,459,859	1,063 MB	2,556	53 s	0.46 GiB	743,808
2^{12} rows (163)	16	77,736,394	1,243 MB	186.9	53 s	0.54 GiB	874,240
2^{16} rows (2,621)	20	88,680,747	1,418 MB	13.32	66 s	0.61 GiB	997,760
2^{18} rows (10,485)	22	**94,069,548**	**1,505 MB**	**3.53**	57 s	0.65 GiB	1,041,024
2^{20} rows (41,943)	24	99,613,898	1,593 MB	0.94	42 s†	0.68 GiB	1,108,992

Table 6: The measured verifier configuration stays below 100M non-free gates and 1.6 GB garbled up to 2^{20} rows, and the garbler needs less than 0.7 GiB. Measured on real Keccak-f proofs; "vars" is the number of packed variables WHIR opens, garbled size is 16 bytes per non-free gate, and η is the garbled size per committed trace cell, 1,625 columns times the rows. Garbling is single-threaded, and its time varied by up to 50% with load on the shared machine (23–35 s at 2^5 across runs). Garbler memory is the plan (one byte per wire) plus 18 bytes per live wire at the peak, computed from the measured counts; at 2^{18} rows the garbler holds at most 2,256,536 labels live. †The 2^{20} proof was generated on a second, larger machine, where the circuit was also garbled, again on one core.

the STARK transcript, the column combination and the AIR constraints, grow with the number of columns, so a narrower AIR shrinks them directly. The largest single item is the ring switch closing: 256 multiplications per packed variable, plus a fixed 11×512 for the column selector.

WHIR alone, and its parameters. Garbling only the WHIR opening of a 18-variable polynomial (rate 1/32, folding 4, 35 + 22 + 16 queries) cost 24.3M non-free gates with single-root commitments, and 21.2M with a Merkle cap of 32 roots. That cap is Plonky3's recommended height and was the optimum in our sweep of 1 to 256 roots. Three quarters of the WHIR circuit was hashing. Across rates 1/16–1/64 and folding factors 2–5, rate 1/32 with folding 4 gave the smallest circuit among the configurations we kept; rate 1/64 saved a further 10% of gates at the price of a prover domain twice as large.

Why a binary field. We also built KoalaBear ($p = 2^{31} - 2^{24} + 1$) and Poseidon2 on wires, to price the prime-field alternative. A KoalaBear multiplication is 3,451 AND gates, and a degree-4 extension multiplication 60,615, against 2,187 for all of $\mathbb{F}_{2^{128}}$. One Poseidon2 permutation of width 16 is 1,240,747 AND gates: 120 Blake3 compressions. The WHIR opening of a 2^{20} KoalaBear configuration (2,157 permutations, about 7,500 extension multiplications) would be about 3.1×10^9 gates with Poseidon2, more than the whole Boolean-garbled Groth16 verifier, and still $0.3\text{–}0.5 \times 10^9$ with Blake3, against our 21M for a 2^{18}-variable opening. The four-fold difference in size does not account for a gap of 15 to 150×: a garbled STARK verifier should use a binary field.

Ablation. Table 8 isolates two of our choices on the measured verifier. Replacing our Blake3 adder with the original two-AND adder, with everything else unchanged, grew the circuit by 32% at 2^5 rows and 29% at 2^8: the adder saves a quarter of the whole verifier, because Blake3 is in the transcript, the leaf hashes and every Merkle path. Building the same 2^5-row verifier as a stored circuit produced exactly the same gates and wires, but took 12.4 GiB and 283 s, against 0.69 GiB and 33 s for the plan and the streamed garbling together, including the garbling itself.

Accounted input-publication components. A Taproot output commits to its tree through a 32-byte output key at funding time; the large cost occurs if the dispute branch is spent, when

phase	non-free gates	share
ring switch closing	31,633,960	33.6%
WHIR Merkle paths	14,613,722	15.5%
column combination	11,582,352	12.3%
AIR constraints (1,650, Horner under α)	10,736,110	11.4%
STARK transcript	10,531,194	11.2%
WHIR leaf hashes	3,320,880	3.5%
WHIR closing query weights	3,069,960	3.3%
WHIR query folds	2,624,400	2.8%
WHIR transcript	1,603,940	1.7%
zerocheck sumcheck	1,227,336	1.3%
ring switch statement	835,688	0.9%
other (9 phases)	2,290,006	2.4%
total	94,069,548	100%

Table 7: A third of the 2^{18}-row verifier is the ring switch closing, and the five largest phases make up 84%. Non-free gates by the verifier phase that emits them.

configuration	non-free gates	peak memory	time
2^5 rows, this work (streamed)	44,573,121	0.69 GiB	33 s (plan + garble)
2^5 rows, stored build	44,573,121	12.4 GiB	283 s (build only)
2^5 rows, original Blake3 adder	58,997,590 (+32%)	–	–
2^8 rows, this work	66,459,859	–	–
2^8 rows, original Blake3 adder	85,768,603 (+29%)	–	–

Table 8: The one-AND adder removes about a quarter of the verifier's gates, and streaming cuts memory 18×. Peak memory is the whole process's, including proof generation, which is negligible at 2^5 rows; the stored build's time excludes garbling.

the witness reveals the authentication signature or preimage, leaf script and control block. Table 9 accounts for that dispute-time input-publication component for the measured 2^{18}-row circuit. The circuit reads 1,041,024 logical bits (127.1 KiB), whereas the serialized proof is 140 KiB. Every authenticated row includes its signature or opening witness, transaction-binding authorization, authentication script and complete reveal-transaction shell. The adaptor and Antichain fixtures use two P2TR outputs per transaction; Lamport uses one. The table therefore compares concrete input-publication components, but not complete or identical dispute graphs.

The executable raw-data fixture provides a transport reference point, not a lower bound. It puts the same logical bits into 1,627 80-byte witness items over two one-leaf P2TR inputs. The serialized transaction includes witness framing, both drop scripts, both 33-byte control blocks, two input shells and one P2TR output; it is 132,788 bytes, 133,193 WU and 33,299 vB, below the 400,000-WU standard-transaction limit [Bit26]. It contains no transaction signature or bit-to-label authentication, uses synthetic outpoints, and has not been submitted to `testmempoolaccept`. It is therefore not an on-chain cost estimate for the protocol. A separate ESSPI-style signed-envelope fixture [LJF25] commits the 130,128-byte payload in a 154-vB wallet-signed transaction and reveals it in a 32,850-vB script-path spend, for 33,004 vB total. Strict Core 31.1 policy accepts the unconfirmed parent–child pair; the reveal includes its transaction signature, 130,917-byte tapscript, control block and transaction shells. It authenticates the bytes but does not deliver the unique selected garbled-circuit labels, and it excludes ESSPI's surrounding data-availability, secondary-

verifier and fraud graph. It is therefore neither an optimized GC Assert nor a total protocol cost.

For one key set, the executable fixtures measure 2.22 MvB with the 8-bit Schnorr-adaptor baseline, 11.85 MvB with safe Antichain Winternitz and 17.27 MvB with Lamport. The adaptor fixture uses the opcode positions required by BIP342 [WNT20], validates every code-separator-specific completed signature and serializes 133 P2TR inputs in 23 two-output transactions. Its largest transaction is 399,936 WU, only 64 WU below the standard-policy limit, so this packing is specific to the measured two-output shell. The Antichain fixture pins the upstream implementation, adds a transaction-binding 64-byte signature per input and serializes 784 inputs in 131 two-output transactions; 11.85 MvB is its all-nonzero worst case. The Lamport fixture serializes every field of 174 one-output transactions: 1,044 P2TR inputs, their 64-byte `SIGHASH_DEFAULT` signatures, preimages, complete leaf scripts and depth-zero control blocks. All selected leaves execute with real BIP341 transaction contexts under Bitcoin Core 31.1 strict policy. The Mosaic source construction makes adaptor publication M-independent through polynomial correlation, but that correlation and its setup checks are not implemented here; the one-key-set fixture uses discrete logarithms. The one-set hash rows require unimplemented soldering; publishing seven independent key sets instead gives the arithmetic 82.94–120.86 MvB projections, without proving consistent bits.

We exported all three transaction-bound fixtures to isolated Bitcoin Core 31.1 regtest nodes with non-standard transactions disabled. Core rejected each representative reveal while its funding parent was unconfirmed as `too-large-cluster`. After funding confirmation, it accepted every reveal. The adaptor fixture used a 5,841-vB funding transaction and 23 reveals totaling 2,222,213 vB, or 2,228,054 vB together, and mined in three blocks. Antichain used 33,836 vB of funding and 131 reveals totaling 11,848,875 vB, or 11,882,711 vB together, and mined in 12 blocks. Lamport used 45,016 vB of funding and 174 reveals totaling 17,265,635 vB, or 17,310,651 vB together, and mined in 18 blocks; 173 reveals are 99,312 vB (397,246 WU) and the tail is 84,659 vB. Core also rejected Lamport's signed 174-parent join while its reveal parents were unconfirmed. After they confirmed, it accepted and mined the 13,060-vB join, bringing that tested slice to 17,323,711 vB. This establishes staged standard-policy feasibility for these bounded signed publication slices, not peer-to-peer propagation, robust fee management, multi-leaf timeout trees or the missing challenge, Disprove, anchor and soldering graph. Adaptor and Antichain join/completion transactions remain unconstructed.

At BABE's historical scenario of 2.2 sat/vB and $95,500 per bitcoin [GKS+26], the one-set measurements have component fee equivalents of 0.049, 0.261 and 0.380 BTC, respectively; these are neither current quotes nor fees for a complete dispute. Their serialized packings use 23, 131 and 174 transactions; dividing by 100 kvB instead gives 23, 119 and 173 capacity equivalents, which are not transaction counts. For the adaptor fixture, minimizing inputs alone uses 131 inputs and 26 transactions (2,222,397 vB); adding two inputs removes three transaction shells and lowers the reveal total by 184 vB. Creating those two extra P2TR prevouts costs about 86 vB, so the latter remains smaller before counting its three fewer completion parents. For example, 1,041,024 Lamport bits require 1,044 inputs at 998 bits per input, and the simplest one-output 400-kWU packing fits at most six full inputs, so the minimal-input, maximal-chunk layout uses 174 transactions. This is not a transaction-count lower bound: repartitioning into more, smaller inputs can fit 173 transactions, but increases the total serialized footprint. The strict Core fixture also shows that mempool policy cannot admit the 174 reveals as a single zero-confirmation graph: funding must confirm first, the independent reveals then require at least 18 blocks, and only then can the join be admitted. The tuned proof of Table 10 reduces input-proportional terms by 18.74%. A deployment must serialize and policy-test the tuned transactions and size its challenge bond to the resulting complete graph.

input mechanism	key sets	vB/bit	accounted component	evidence and accounting boundary
raw transport baseline	0	0.032	33.30 kvB	serialized two-input fixture: data, drop scripts, control blocks and tx shell; no signature or label authentication
signed envelope transport [LJF25]	0	0.032	33.00 kvB	Core-tested commit and signed reveal; authenticates bytes but supplies no selected GC labels and excludes the DA/fraud graph
Schnorr adaptor, 8-bit baseline [WAA+26, KTA+26]	1	2.13	2.22 MvB	Core-tested 133-input, 23-transaction fixture; completed signatures, code separators, scripts, depth-zero controls and two P2TR outputs
Antichain $(2, 16)$ [TF26]	1	11.38	11.85 MvB	Core-tested all-nonzero worst case: 784 signed inputs in 131 two-output transactions using pinned upstream openings and scripts
Lamport	1	16.59	17.27 MvB	fully serialized 174-reveal fixture: transaction-binding signatures, preimages, scripts, depth-zero controls, input/output shells; Core 31.1 accepted after funding confirmation
Antichain, independent keys	7	79.67	82.94 MvB	arithmetic 7× worst-case projection; consistency and graph unimplemented
Lamport, independent keys	7	116.10	120.86 MvB	arithmetic 7× projection; consistency and graph unimplemented

Table 9: Source-specific accounting of the dispute-time circuit-input publication component for 1,041,024 logical bits. The adaptor row includes completed transaction-binding Schnorr signatures. Each hash-based row includes its one-time opening witness plus an ordinary transaction-binding Schnorr signature. Every label-authentication row also includes the authentication scripts, control blocks, input/output shells and reveal-transaction framing. Adaptor and Antichain use two outputs per transaction, whereas Lamport uses one. Funding for all three and Lamport completion are reported separately in the text. The complete challenge/timeout/Disprove graph, anchors, robust fee management and multi-copy soldering are not included. The table therefore does not report total on-chain cost.

How far the input shrinks. We counted the verifier's input bits by proof component from the WHIR schedule alone. The count equals the measured input count at the five tested configurations through 2^{18} rows. We then swept rates 1/8 to 1/256, folding factors 2 to 6 and grinding budgets up to 48 bits at 2^{18} rows, keeping 110 bits per WHIR term in the Johnson regime (Table 10). The smallest input, at rate 1/256, folding 4 and 48 bits of grinding, is 845,952 bits, 19% below the measured configuration, at 103.9 bits of composed soundness with every term assessed. Its WHIR part shrinks by 35%, but 480,256 bits do not depend on WHIR at all: the opened values are 416,000 bits, two rows of 1,625 columns at 128 bits each. The tuning also moves soundness from queries to grinding, which halves under quantum search (§5.1). Below this the input shrinks only with a narrower AIR, whose opened values scale with its width, or by recursion into a proof whose own WHIR opening is still some 0.37M bits at these parameters; a measured recursion reaches 564,552 bits (Table 11).

Narrow statements and batching. Holding the height fixed while narrowing the AIR discards most of the represented computation, so we instead required at least as many committed cells as the measured $1{,}625 \times 2^{18}$ trace and rounded the height up to a power of two. Under the current WHIR profile, hypothetical widths 128 and 32 at heights 2^{22} and 2^{24} require 659,840 and 636,288 input bits, reductions of 36.6% and 38.9%. Combining those layouts with the best configuration in the rate/folding/48-bit-grinding sweep gives 464,768 and 441,216 bits, reductions of 55.4% and 57.6%. These are exact schedule counts for layout models, not generated proofs: no such narrow Keccak AIR is implemented, the power-of-two rounding represents 126.03% of the original cell count, and the aggressive result inherits the quantum-search and prover-cost caveats of grinding.

component	measured	tuned
opened values (current and next row)	416,000	416,000
zerocheck and ring switch	64,256	64,256
WHIR Merkle paths	346,624	239,616
WHIR leaf rows	163,840	92,160
WHIR caps, OOD answers, sumchecks, final polynomial	50,304	33,920
total input bits	1,041,024	845,952
STIR queries per round	$33 + 20 + 15 + 12$	$16 + 12 + 9 + 8$
rate, folding, grinding budget	1/32, 4, 32	1/256, 4, 48
composed soundness (bits)	104.3	103.9

Table 10: Tuning WHIR removes a fifth of the input, and the AIR's opened values set the floor. Input bits at 2^{18} rows by proof component, for the measured configuration and the smallest configuration found in the sweep; 128 bits per field element and 256 per digest, with full Merkle paths below each cap.

Batching has stronger evidence because both endpoints are real artifacts. The 2^{18} and 2^{20} traces contain 10,485 and 41,943 Keccak-f permutations, while their verifier inputs are 1,041,024 and 1,108,992 bits. Thus 4.0003× as much work increases the absolute input by only 6.53%, reducing the input per permutation from 99.29 to 26.44 bits, a 3.76× amortization. Batching does not reduce the absolute unhappy-path payload for one proof, but it is the only measured input optimization here that requires no new AIR or input-delivery protocol.

Recursion into a narrow machine. The floor above belongs to the statement, so we also garbled the verifier of a statement built to be verified cheaply. Ziren's binary stage [ZKM26] proves, over $\mathbb{F}_{2^{128}}$ with Boolean WHIR and Blake3, the program that verifies a shrunk zkVM proof, and its `binary-recursion` crate records Plonky3's verifier of such a proof as a straight-line tape over $\mathbb{F}_{2^{128}}$. A five-table tape machine (ledger, arithmetic, rewiring, Blake3 message and round tables) proves a run of a tape, and its own verifier is recorded and proved again: narrow recursion. We translate a tape gate by gate (`whir-gc/src/binary_tape.rs`), trusting Ziren's recorder to capture Plonky3's verifier. The tape machine binds no public values, so from level 2 on the binary stage's public-value digest, which commits to the zkVM program's verifying key and outputs, is a private witness. The circuit therefore attests that some zkVM proof verifies, not which program ran or what it output. For a Fibonacci program the binary stage's verifier has 3.37G non-free gates and 18,050,304 input bits; after three rounds of narrow recursion the verifier of the last proof has the sizes in Table 11. Every row is the verifier of a real proof: its circuit accepts that proof with every tape value reproduced and rejects three single-input changes. From the third row on, each is a level-4 proof re-proved from the same level-3 tape at the Johnson rate 1/1024 and re-recorded, and each was garbled by the streaming garbler. The last has 564,552 input bits (9,513 bytes and 3,816 field elements), 0.54× the Keccak verifier's, in 218.0M non-free gates and 3.49 GB.

The steps combine recording choices with changes to the proof system. The recorder reads opened values the verifier knows (padding and constant preprocessed columns) as constants, and reads the fixed preprocessed Merkle tree down to 18 layers below its root as circuit constants, chosen by a one-hot of the query's top index bits. The schedule folds 4, 5, 4, 4 variables per WHIR round at 46 bits of grinding. The protocol changes are five. Ring-switch claims at points sharing the coordinates one packed element absorbs send one tensor, weighted by powers of a challenge drawn after the points, and a pair's main and preprocessed switches run as one over both packings

verifier	input bits	non-free gates	garbled
binary stage (level 1)	18,050,304	3,371.4M	53.94 GB
narrow recursion, level 2, Ziren `9398469a`	2,356,864	473.9M	7.58 GB
level 4, rate 1/1024, 64-byte rewiring rows	1,248,384	231.2M	3.70 GB
+ known opened values as constants	1,137,536	226.8M	3.63 GB
+ one tensor for claims sharing a point	1,006,464	221.7M	3.55 GB
+ 200-bit Merkle digests	895,504	220.2M	3.52 GB
+ packed operand columns	834,832	219.2M	3.51 GB
+ linear forms, per-round folds, deep preprocessed cap	643,688	215.8M	3.45 GB
+ one ring switch per pair, eq-factored bus rounds	603,112	216.5M	3.46 GB
+ affine preprocessed forms, one shared forms sumcheck	574,952	217.8M	3.48 GB
+ preprocessed cap 18 layers deep	**564,552**	218.0M	3.49 GB

Table 11: Narrow recursion removes the statement's width from the garbled verifier's input. Exact counts for Ziren's binary stage and for the verifier of its narrow-recursion proofs (level 2 with 26-bit addresses, the smallest that fit the real tape); the rows below the third use the experimental changes listed in the text.

behind a selector variable. Merkle digests are Blake3 truncated to 200 bits, which keeps collisions at 2^{100} work classically and $2^{66.7}$ under BHT search. Each table sends one value per linear form of its columns that its constraints and buses read, derived from the symbolic AIR, and a sumcheck over the column index reduces them to the table's ring-switch claim; the verifier rebuilds the columns with a constant right inverse, which costs only XORs. Preprocessed forms are cut to those independent over the fixed trace's GF(2) column basis and the all-ones column, the rest rebuilt by an affine inverse, and one sumcheck over the widest table's column index serves every table. Bus product-GKR rounds send the round polynomial without its linear eq factor. Each change passes Plonky3's and Ziren's tests and tamper tests of its own, but none has been reviewed; the patches are in the repository. Each proof meets 100 bits of composed soundness by Plonky3's assessment in the Johnson regime, with up to 46 bits of grinding per term at level 4, which halves under quantum search. The circuit checks only the last proof, so the statement's soundness also composes over the earlier levels. Proving level 4 alone took 1.8 hours on 64 cores with four GPUs grinding and about 470 GB of memory. Serializing the input-publication fixtures of Table 9 at 564,552 bits, with every transaction signed and every leaf executed under the stack limit, gives 1,205,217 vB for the 8-bit adaptor baseline (73 inputs in 13 reveal transactions), 6,425,757 vB for Antichain $(2, 16)$ (426 inputs in 71) and 9,363,254 vB for Lamport (566 inputs in 95), plus a 7,155 vB Lamport join; Disprove is one 97 vB transaction. Plain 4-bit Winternitz, modeled from one executed input at 23.06 weight units per bit, would be about 3.26 MvB, but it cannot deliver garbled labels without a translation layer. These reveals were not replayed in Bitcoin Core, and the accounting boundary is that of Table 9.

Comparison. Table 12 places the result among approaches to verifying proofs on Bitcoin (§8). For the designated-verifier pairing SNARK we garbled bitvm-gc's circuit with our streaming garbler. It has 481.6M non-free gates and 1.92×10^9 wires, with at most 15.6M live, and used 2.2 GB peak working memory in under six minutes. The circuit accepted the valid proof and rejected it with one witness bit flipped; built as a stored circuit, the same verifier needed more than 64 GB. The Keccak verifier has 5.1× fewer non-free gates, and 30× fewer than the Boolean-garbled Groth16 verifier; the experimental recursion verifier (Table 11) has 2.2× and about 14× fewer. Native-group garbling of Groth16 [KFLT26] and witness encryption [GKS+26] produce far smaller artifacts, of megabytes,

and their proofs need far fewer input bits; both rely on pairings and a trusted setup. On-chain, the comparison inverts. The Groth16 designs publish 508 to 1,019 input bits and report dispute paths of a few to a few tens of kilobytes, under $60. Our Keccak dispute publishes 1,041,024 bits, and its Core-tested input component alone is 2.22 MvB with adaptor signatures or 11.85 MvB hash-based; the recursion verifier's 564,552 bits serialize to 1.21 and 6.43 MvB, not replayed in Bitcoin Core. That is one to three orders of magnitude more (about 38× deferred binding's 32 kvB even for the recursion verifier with adaptors), before the unconstructed remainder of the graph. That is the price of verifying the STARK directly rather than wrapping it, and §6 and §9 are about reducing it.

approach	statement	off-chain size	input bits	on-chain dispute path, as reported	assumptions
Boolean garbled verifier [Bit25]	Groth16	42 GiB	1,019	Assert 2.4 kvB and Disprove 93 vB, about $5 at 2 sat/vB [WAA+26]; the published experiment normalized to $37.65 [GKS+26]	pairings, trusted setup
native-group garbling [KFLT26]	Groth16	2.4 MiB	n/r	n/r	pairings, trusted setup
garbled DV-Pari (bitvm-gc, our measurement)	DV-SNARK	7.7 GB	n/r	n/r	elliptic curves, secret trapdoor
BABE [GKS+26]	Groth16	22.2 MiB	508	Assert 9,240 vB, ChallengeAssert 17,400 vB, WronglyChallenged 149 vB: 26.8 kvB, $56.90 at 2.2 sat/vB	pairings (GGM+RO), trusted setup
deferred-binding WE [Pha26]	Groth16, late inputs	0.92 GiB†	768	Assert 2,154 B, ChallengeAssert 14,312 B witness and 109,393 B script, WronglyChallenged 64 B; about 32 kvB	BABE + DLog
PIPE with AADP WE [ACG+26, SRG+26]	pairing SNARK	≈ 338 TB (estimate)	one hash	one hash in Assert, one signature in Disprove; no size reported	heuristic
this work	**Keccak-f STARK over $\mathbb{F}_{2^{128}}$, 2^{18} rows**	**1.5 GB**	**1,041,024**	**input publication 2.22 MvB (adaptor, DLog) or 11.85 MvB (hash-based), about $4,700 or $24,900 at 2.2 sat/vB, Core-replayed; Disprove 97 vB; graph incomplete§**	**Blake3/RO; assumed CCR and adaptive authenticity‡**
this work, narrow recursion (experimental, not Core-replayed)	STARK recursion proof over $\mathbb{F}_{2^{128}}$	3.49 GB	564,552	input publication 1.21 MvB (adaptor, DLog) or 6.43 MvB (hash-based), about $2,500 or $13,500 at 2.2 sat/vB; Disprove 97 vB; graph incomplete§	as above, plus unreviewed proof-system changes‡

Table 12: Comparison of verifier artifacts, dispute inputs and on-chain dispute cost. Off-chain artifact per instance, in bytes (garbled circuits at 16 bytes per non-free gate); the number of input bits published on-chain at dispute time; and the on-chain dispute path, each as its source reports it; n/r: not reported. †One finalized instance: 59.3M non-free gates across three garbled circuits ($r\pi_1$, $Q = x_D L_D + B$ and rQ) plus adaptor tables, as the implementation report measures it; a 3.83 GiB circuit artifact is shared by all instances. Its on-chain entry is witness and script bytes as reported; the kvB figure is our arithmetic. §For one key set. The Keccak entry is the Core-tested reveals of Table 9. The recursion entry serializes the same fixtures at 564,552 bits, with every transaction signed and every leaf executed under the stack limit, but was not replayed in Bitcoin Core (§7). Both add the Core-tested reject-label Disprove; the challenge, completion, timeout and anchor transactions and the multi-copy soldering are not constructed (§6), so each is a lower bound on the dispute path, not a total. ‡Verifier assumptions only. A surrounding protocol additionally uses discrete-logarithm signatures, or requires the unimplemented hash-based multi-copy soldering and a post-quantum transaction-authorization mechanism. Neither statement has public values; the recursion row binds neither the zkVM program nor its outputs. The recursion row is the verifier of the last narrow-recursion proof of Ziren's binary stage (Table 11), with the experimental proof-system changes listed in §7.

8 Related work

We group prior work by how it verifies a proof on Bitcoin: by garbling a verifier, by witness encryption, or through small pairing-based SNARKs that both approaches rely on. We close with the garbling techniques we use.

BitVM and garbled verifiers. BitVM2 [LAA+25] verified optimistically by bisecting the verifier's trace on-chain. BitVM3 garbled the verifier instead [Lin25, Che25], and its formal treatment defines BitVM3-CORE as a sound and complete on-chain proof system with on-chain costs of about $9 [WAA+26]. The Boolean-garbled Groth16 verifier [Bit25] has about 3×10^9 non-free gates, 42 GiB, with cut-and-choose setup of over three hours at $(N, M) = (181, 7)$ [GKS+26]. Garbling Groth16 with native elliptic-curve group operations reduced the garbled program to 2.4 MiB [KFLT26]. Glock [Eag25] formalized garbled locks and garbled a designated-verifier variant of Pari; BABE reports its garbled circuit at about 12 MB on a binary elliptic curve [GKS+26]. GOAT's bitvm-gc [GOA26b] garbles Groth16 and a designated-verifier form of Pari [DMS24] over BN254 and over the binary curve sect233k1. We garble a measured STARK-verifier path with the same garbling code, and we contribute our streaming garbler and Blake3 circuit back to it. BitVM3's comparison of garbling schemes already lists a Yao-style garbled STARK verifier, at about 10 GB of off-chain storage and as an estimate [WAA+26, Table 9]; we build one and measure 1.5 GB per instance. TRAPGC-DV [FBF+26] verifies Groth16 by designated evaluation: the prover wraps its proof in an outer Groth16 proof under a per-session trapdoor SRS and garbles a small circuit that checks it in scalar form, with cut-and-choose and a universal-composability analysis. It too rests on pairings.

Garbling on Bitcoin: security and inputs. Futoransky, Larotonda and Barbara [FLB25] broke the authenticity of the RSA-based garbling proposed for BitVM3 [Lin25], of the label forward-propagation variant [FDZ25], and of linear adaptors: coprime public exponents let the evaluator forge labels by extended Euclid, and linear adaptors fall to the Franklin–Reiter related-message attack. Our hash-based garbling has no such label algebra. Mosaic [KTA+26] made cut-and-choose maliciously secure with an on-chain footprint independent of the number of garbled copies, and Antichain Winternitz [TF26] published input labels on-chain at a cost up to 52.9% lower than Lamport signatures. Both are independent of the garbled circuit; §6 prices both for our million-bit input.

Quantum-safe transaction authorization. QSB [Lev26] constructs a hash-based authorization path under existing legacy-Script consensus rules by combining HORS-style commitments with transaction-bound hash-to-signature grinding. A QSB spend was mined on Bitcoin mainnet in August 2026 [Sta26]. The construction is non-standard and requires direct miner inclusion. It authorizes a specially constructed spend; adapting it to a multi-transaction, pre-signed and time-locked dispute graph remains open.

Verifying a STARK in Script. Our companion work, in an earlier version of this paper's repository, verifies the same kind of proof directly in Bitcoin Script, with Poseidon2 over KoalaBear and WHIR and without `OP_CAT` [GOA26a]. It needs no garbling and no trusted party, but its 2^{20} example configuration requires 1.24 GB of script, equivalent to 309 blocks; moreover, a standard transaction holds less than one Poseidon2 permutation, so it must be split into sub-query steps. Garbling moves that cost off-chain.

Witness encryption on Bitcoin. BABE [GKS+26] replaced the garbled Groth16 verifier with witness encryption for linear pairing relations [GKPW24]. It made the one quadratic pairing $e(\pi_1, \pi_2)$ linear by computing $r\pi_1$ in a small garbled circuit, built from decomposable randomized encodings on top of Argo MAC [EL26]. The per-instance artifact is 22 MiB, the on-chain cost is comparable to BitVM3, and cut-and-choose setup takes seconds. Deferred binding [Pha26] extended BABE to public inputs known only at dispute time: a second garbled circuit computes a blinded point $Q = x_D L_D + B$ from the late input and a third computes rQ, relying on Groth16's linear public-input aggregation. Its implementation reports 0.92 GiB per finalized instance, 768 input labels, and an Assert of 2,154 bytes with a 14,312-byte ChallengeAssert witness.

Witness encryption for covenants. Bitcoin PIPEs v2 [ACG+26] witness-encrypt a Schnorr signing key under an NP statement to emulate covenants without a soft fork. They define witness signatures, and they map BitVM's disprove condition to a relation $f(w) = 0 \wedge h = \mathrm{Hash}(w)$, which requires the witness at setup. Their witness encryption comes from arithmetic affine determinant programs [SRG+26], an arithmetic variant of affine determinant programs [BIJ+20]. The ciphertext grows as $v(2g+1)^2$ field elements, cubic in the constraint count, which is about 338 TB for a purpose-built 14,371-constraint pairing-SNARK verifier. Security is heuristic: an earlier version fell to a rank-drop attack, and the linearization analysis is described as inconclusive. All witness-encryption designs above are smaller off-chain than a garbled verifier, but all require the statement to be a pairing SNARK, so a STARK would first have to be wrapped in one, and none is post-quantum.

Small SNARK verifiers. Pari [DMS24] has the smallest known pairing-based proofs, two $\mathbb{G}_1$ and two field elements, with a three-pairing verifier. Its equifficient polynomial commitments also give Garuda, a prover with free linear gates. Pari underlies bitvm-gc's designated-verifier SNARK, Glock, and the SNARK of the AADP construction; Glock and bitvm-gc's `sect233k1` verifier use its designated-verifier form on a binary curve, which needs no pairing but still rests on discrete logarithms. Groth16 [Gro16] remains the deployed baseline.

Binary-field proof systems. Binius [DP25] builds succinct arguments over towers of binary fields, and ring switching [DP26] turns a claim about a polynomial's bits into one about its packed extension-field form; Plonky3's Boolean WHIR commitment, which we garble, uses both ideas with WHIR [ACFY25] as the proximity test. Its cheap additions are what make a garbled verifier small.

Garbling. We use privacy-free garbling [FNO15] with free-XOR [KS08], one ciphertext per AND gate [ZRE15], and the syntax of Bellare, Hoang and Rogaway [BHR12b]; free-XOR's security rests on circular correlation robustness [CKKZ12]. Witness encryption in general goes back to Garg, Gentry, Sahai and Waters [GGSW13].

9 Discussion

Why the operator garbles. The roles could be swapped, with a challenger garbling and the operator evaluating its own proof to obtain an accept label that unlocks its payment. The operator then needs the input labels of its proof from the garbler. Publishing both labels of every input wire would disclose Δ, their XOR, and with it every label; handing them over on request lets the garbler withhold them, so that the operator's payment depends on the challenger's cooperation. With the operator as garbler, a withheld label only hurts the party that withholds it.

Input authentication dominates the measured components. The measured verifier and one-copy Script components use primitives available today, but we have not shown that the complete multi-transaction protocol is enforceable under current relay policy. The three signed funding–reveal slices are enforceable only in confirmed stages under Core 31.1; none is an atomic unconfirmed graph. Only the Lamport fixture also constructs and tests a join. The circuit input has 1,041,024 bits (127.1 KiB), distinct from the 140 KiB serialized proof. The accounted dispute-time authentication component consists of the signature or opening witness, its verification script and the row-specific transaction envelope; it is 2.22–17.27 MvB for one key set, with 2.22 MvB denoting the 8-bit adaptor baseline (Table 9). The reveal fixtures have explicit but not identical output shapes and are not total on-chain costs. The hash-based one-set figures require unimplemented multi-copy soldering, and the independent $M = 7$ component projection is 82.94–120.86 MvB before the missing graph. Among the components measured here, the raw bits fit either the unauthenticated 33.3-kvB fixture or a 33.0-kvB signed envelope, but neither supplies selected GC labels. Label-authenticated reveal rather than byte transport or the Taproot commitment itself is therefore the bottleneck. WHIR tuning removes only a fifth of the input (Table 10); equal-area narrow-statement models remove 37–58%, but require a new AIR, and the larger real trace shows that batching can amortize input per permutation by 3.76× without reducing the absolute logical circuit input. Narrow recursion reaches 0.54× the input with real proofs (Table 11), at the price of a recursion prover and of proof-system changes still to be reviewed. Deployability additionally requires public-input support and a concrete, serialized and policy-tested transaction graph.

Conditional input compression. TinyLabels compresses honest selected-label delivery to $|x| + \mathrm{poly}(\lambda)$ online bits under Ring-LWE in the random-oracle model [DLL24]. Its reported concrete batch supports 699,050 128-bit labels with an approximately 56-KB selector key, so the present input would need two reported-size batches: roughly 0.24 MB for the selector keys and bit vector before Bitcoin encoding. This is a useful communication frontier, not a drop-in replacement for Table 9. The selector key is not publicly verifiable, and the malicious-2PC wrapper turns malformed delivery into an abort rather than public evidence. A malicious or withheld key can therefore leave the evaluator with garbage labels but no reject label that opens Disprove; Bitcoin Script cannot cheaply check the underlying Ring-LWE relation. A publicly verifiable or penalizable Batch-Select, including consistency across the M kept garblings, remains an open input-layer direction.

Segmented verification. Another architecture would commit once to a fixed serialization of the proof data, locate a faulty verifier segment by bisection and authenticate only the challenged chunk. For $N = 1{,}041{,}024$ bits split into chunks of at most B bits under a 256-bit Merkle tree, its data-only challenged input is

$$B + 256\,\lceil \log_2 \lceil N/B \rceil \rceil + C_{\mathrm{state}},$$

where C_{state} binds the transcript, accumulator and boundary state. For example, a 6,656-bit first-round query chunk has 157 chunks and an eight-digest path, hence 8,704 bits before state. Query segmentation alone is insufficient: leaf rows and paths account for 510,464 input bits, leaving a 530,560-bit global remainder containing the opened rows, sumchecks, ring switch and transcript. Achieving the bound requires segmenting those stages, binding their boundary states and constructing a fault-location and withholding penalty protocol. None is implemented here; without those mechanisms the worst-case disclosure remains the full N bits.

Other AIRs. Our sizes are for the 1,625-column Keccak AIR. Plonky3 also has binary AIRs for SHA-256 and Blake3, but they read no next-row columns, which our verifier currently assumes;

supporting them changes the opening layout and is left to future work. The column-dependent terms, about a third of the circuit, scale with the AIR's width.

Verifying the garbling. A garbler must be kept honest, and in current deployments that is cut-and-choose over many instances. Its cost is linear in the garbled circuit. At $(N, M) = (181, 7)$ and 2^{18} rows, the operator garbles 181 instances at about 57 s each, 2.9 core-hours or about 11 minutes on 16 cores, and a challenger stores the 7 kept instances, 10.5 GB. BitVM3 reports 3 hours 24 minutes on 16 cores and 288.4 GB for the Boolean-garbled Groth16 verifier at the same parameters [WAA+26]. Against native-group garbling of Groth16 [KFLT26] or BABE, whose artifacts are megabytes, our setup is larger. The streaming garbler re-garbles an opened instance from its seed in one garbling pass, reusing the plan that all instances share. These setup figures are derived from the measured garbling time, not measured end to end.

Further reductions. The ring switch closing is a third of the circuit and is the first target. An $\mathbb{F}_{2^{128}}$ multiplier below Karatsuba's 3^7 is a second: bitvm-gc's $\mathbb{F}_{2^{233}}$ multiplier, which evaluates, multiplies and interpolates over $\mathbb{F}_{2^9}$, costs 2,097 AND gates for a field almost twice as wide, at the price of 1.5M XORs per product. Per-tree Merkle cap heights and a narrower AIR for the statement actually verified reduce the hashing and column terms; the recursion of Table 11 does both, with a preprocessed tree read deep below its root and tables whose opened values are linear forms of their columns. At the other end of the graph, a Core-tested false-label-derived Taproot pointlock reduces this implementation's 16-byte-label Disprove spend from 97 to 83 vB when both outputs retain the same timeout sibling. This is only a 14-vB micro-optimization: it adds the Schnorr/discrete-log assumption, does not reveal the false label and cannot by itself constrain the payout outputs.

10 Conclusion

We have shown that the verification path exercised by one no-public-input Keccak-f configuration of Plonky3's Boolean-WHIR pipeline fits garbling well. At 2^{18} rows it has 94.1M non-free gates and a 1.5 GB garbled artifact, about 30× fewer non-free gates than the reported Boolean-garbled Groth16 verifier. This establishes the verifier-engineering result, not an end-to-end Bitcoin protocol.

The largest measured on-chain component is revealing and authenticating the 1,041,024 circuit-input bits in a dispute, including their signatures or preimages and verification scripts, rather than committing a Taproot output or transporting the 127.1-KiB raw encoding. The executed components accounted here span 2.22–17.27 MvB for one input-key set, where 2.22 MvB is the 8-bit adaptor baseline, with explicit but different output shapes. Strict Core 31.1 accepted their staged funding–reveal slices, which total 2.228, 11.883 and 17.311 MvB. Lamport's additional tested join raises its bounded slice to 17.324 MvB, but the total protocol footprint remains unmeasured. Hash-based multi-copy soldering is unimplemented, and the independent seven-set component projection is 82.94–120.86 MvB before the transaction graph. The approximately 52-bit quantum figure applies to the measured proof-system parameters under the stated assumptions, while cut-and-choose is configured at 40 statistical bits and the current transaction authorization is not post-quantum. Before deployment, the construction needs public-input support, an implemented and measured multi-copy input layer, a serialized and policy-tested Assert graph, post-quantum graph authorization, and a measured parameter set with an enlarged field and labels, a wider-state hash and stronger cut-and-choose.

The input shrinks further when the statement is itself a recursion proof. Verifying the last of three narrow-recursion proofs of Ziren's binary stage takes 564,552 input bits and 218.0M non-free

gates, measured on real proofs but resting on experimental, unreviewed changes to Plonky3 and Ziren: shared ring-switch tensors, 200-bit digests, linear-form openings with affine preprocessed forms and eq-factored bus rounds.

References

[ACFY24] Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev. STIR: Reed–solomon proximity testing with fewer queries. In *Advances in Cryptology – CRYPTO 2024, Part X*, volume 14929 of *LNCS*, pages 380–413. Springer, 2024. Full version: Cryptology ePrint Archive, Paper 2024/390.

[ACFY25] Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev. WHIR: Reed–Solomon proximity testing with super-fast verification. In *Advances in Cryptology – EUROCRYPT 2025, Part IV*, volume 15604 of *LNCS*, pages 214–243. Springer, 2025. Cryptology ePrint Archive, Paper 2024/1586.

[ACG+26] Michel Abdalla, Brent Carmer, Muhammed El Gebali, Handan Kilinc-Alper, Mikhail Komarov, Yaroslav Rebenko, Lev Soukhanov, Erkan Tairi, Elena Tatuzova, and Patrick Towa. Bitcoin PIPEs v2: Covenants and ZKPs on Bitcoin via witness encryption. Cryptology ePrint Archive, Paper 2026/186, 2026.

[BHR12a] Mihir Bellare, Viet Tung Hoang, and Phillip Rogaway. Adaptively secure garbling with applications to one-time programs and secure outsourcing. In *Advances in Cryptology – ASIACRYPT 2012*, volume 7658 of *LNCS*, pages 134–153. Springer, 2012.

[BHR12b] Mihir Bellare, Viet Tung Hoang, and Phillip Rogaway. Foundations of garbled circuits. In *Proceedings of the 2012 ACM Conference on Computer and Communications Security*, pages 784–796. ACM, 2012.

[BHT98] Gilles Brassard, Peter Høyer, and Alain Tapp. Quantum cryptanalysis of hash and claw-free functions. In *LATIN 1998: Theoretical Informatics*, volume 1380 of *LNCS*, pages 163–169. Springer, 1998.

[BIJ+20] James Bartusek, Yuval Ishai, Aayush Jain, Fermi Ma, Amit Sahai, and Mark Zhandry. Affine determinant programs: A framework for obfuscation and witness encryption. In *11th Innovations in Theoretical Computer Science Conference (ITCS 2020)*, volume 151 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 82:1–82:39. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2020.

[Bit25] BitVM. Garbled SNARK verifier. Source code, commit `5a2cd4dc6cb19e37adb1b3ab94414e01d1e8b338`, 2025.

[Bit26] Bitcoin Core developers. Bitcoin Core 31.1 transaction relay policy constants. Source code, release tag `v31.1`, 2026. Defines `MAX_STANDARD_TX_WEIGHT` and `DEFAULT_CLUSTER_SIZE_LIMIT_KVB`.

[Can89] David G. Cantor. On arithmetical algorithms over finite fields. *Journal of Combinatorial Theory, Series A*, 50(2):285–300, 1989.

[CBBZ23] Binyi Chen, Benedikt Bünz, Dan Boneh, and Zhenfei Zhang. HyperPlonk: Plonk with linear-time prover and high-degree custom gates. In *Advances in Cryptology – EUROCRYPT 2023, Part II*, volume 14005 of *LNCS*, pages 499–530. Springer, 2023. Full version: Cryptology ePrint Archive, Paper 2022/1355.

[Che25] Weikeng Chen. SoK: BitVM with succinct on-chain cost. Cryptology ePrint Archive, Paper 2025/1253, 2025.

[CKKZ12] Seung Geol Choi, Jonathan Katz, Ranjit Kumaresan, and Hong-Sheng Zhou. On the security of the "free-XOR" technique. In *Theory of Cryptography Conference (TCC 2012)*, volume 7194 of *LNCS*, pages 39–53. Springer, 2012.

[CMS19] Alessandro Chiesa, Peter Manohar, and Nicholas Spooner. Succinct arguments in the quantum random oracle model. In *Theory of Cryptography Conference (TCC 2019), Part II*, volume 11892 of *LNCS*, pages 1–29. Springer, 2019. Cryptology ePrint Archive, Paper 2019/834.

[DLL24] Marian Dietz, Hanjun Li, and Huijia Lin. TinyLabels: How to compress garbled circuit input labels, efficiently. Cryptology ePrint Archive, Paper 2024/2048, 2024.

[DMS24] Michel Dellepere, Pratyush Mishra, and Alireza Shirzad. Garuda and pari: Faster and smaller SNARKs via equifficient polynomial commitments. Cryptology ePrint Archive, Paper 2024/1245, 2024.

[DP25] Benjamin E. Diamond and Jim Posen. Succinct arguments over towers of binary fields. In *Advances in Cryptology – EUROCRYPT 2025, Part IV*, volume 15604 of *LNCS*, pages 93–122. Springer, 2025. Full version: Cryptology ePrint Archive, Paper 2023/1784.

[DP26] Benjamin E. Diamond and Jim Posen. Polylogarithmic proofs for multilinears over binary towers. In *Advances in Cryptology – EUROCRYPT 2026, Part VII*, volume 16547 of *LNCS*, pages 3–32. Springer, 2026. Full version: Cryptology ePrint Archive, Paper 2024/504.

[Eag25] Liam Eagen. Glock: Garbled locks for Bitcoin. Cryptology ePrint Archive, Paper 2025/1485, 2025.

[EL26] Liam Eagen and Ying Tong Lai. Argo MAC: Garbling with elliptic curve MACs. Cryptology ePrint Archive, Paper 2026/049, 2026.

[FBF+26] Ariel Futoransky, Fadi Barbàra, Ramses Fernandez, Gabriel Larotonda, and Sergio Demian Lerner. TRAPGC-DV: Trapdoor garbled circuit designated verification of Groth16 on Bitcoin. Cryptology ePrint Archive, Paper 2026/1430, 2026.

[FDZ25] Alva Fu, Stephen Duan, and Ethan Zhu. Instantiating BitVM3 from label forward propagation. `https://www.goat.network/bitvm3-label-forward-propagation`, 2025.

[FLB25] Ariel Futoransky, Gabriel Larotonda, and Fadi Barbara. A note on the security of the BitVM3 garbling scheme. Cryptology ePrint Archive, Paper 2025/1291, 2025.

[FNO15] Tore Kasper Frederiksen, Jesper Buus Nielsen, and Claudio Orlandi. Privacy-free garbled circuits with applications to efficient zero-knowledge. In *Advances in Cryptology – EUROCRYPT 2015, Part II*, volume 9057 of *LNCS*, pages 191–219. Springer, 2015.

[GGSW13] Sanjam Garg, Craig Gentry, Amit Sahai, and Brent Waters. Witness encryption and its applications. In *45th ACM STOC*, pages 467–476. ACM, 2013.

[GKPW24] Sanjam Garg, Dimitris Kolonelos, Guru-Vamsi Policharla, and Mingyuan Wang. Threshold encryption with silent setup. In *Advances in Cryptology – CRYPTO 2024, Part VII*, volume 14926 of *LNCS*, pages 352–386. Springer, 2024.

[GKS+26] Sanjam Garg, Dimitris Kolonelos, Mikhail Sergeevitch, Srivatsan Sridhar, and David Tse. BABE: Verifying proofs on Bitcoin made 1000x cheaper. Cryptology ePrint Archive, Paper 2026/065, 2026.

[GOA26a] GOAT Network. bitcoin-stark-verifier: a STARK verifier in Bitcoin Script without OP_CAT. GOATNetwork/bitcoin-stark-verifier, crates `poseidon2` and `whir`, commit `d45e453`, 2026.

[GOA26b] GOAT Network. bitvm-gc: garbled SNARK verifier and BABE for BitVM. Source code, commit `10083d4a63eb08a344dc40ee84bc11394b218a89`, 2026.

[Gro96] Lov K. Grover. A fast quantum mechanical algorithm for database search. In *Proceedings of the 28th Annual ACM Symposium on Theory of Computing*, pages 212–219. ACM, 1996.

[Gro16] Jens Groth. On the size of pairing-based non-interactive arguments. In *Advances in Cryptology – EUROCRYPT 2016, Part II*, volume 9666 of *LNCS*, pages 305–326. Springer, 2016.

[KFLT26] Nakul Khambhati, Aaron Feickert, Christian Lewe, and Mukesh Tiwari. Garbling Groth16 with native group operations. Cryptology ePrint Archive, Paper 2026/2100, 2026.

[KS08] Vladimir Kolesnikov and Thomas Schneider. Improved garbled circuit: Free XOR gates and applications. In *Automata, Languages and Programming (ICALP 2008), Part II*, volume 5126 of *LNCS*, pages 486–498. Springer, 2008.

[KTA+26] Nakul Khambhati, Mukesh Tiwari, Azz, Sapin Bajracharya, Manish Bista, Liam Eagen, Christian Lewe, and Aaron Feickert. Mosaic: Practical malicious security for garbled circuits on Bitcoin. Cryptology ePrint Archive, Paper 2026/812, 2026.

[LAA+25] Robin Linus, Lukas Aumayr, Zeta Avarikioti, Matteo Maffei, Andrea Pelosi, Orfeas Stefanos Thyfronitis Litos, Christos Stefo, David Tse, and Alexei Zamyatin. Bridging Bitcoin to second layers via BitVM2. Cryptology ePrint Archive, Paper 2025/1158, 2025.

[LCH14] Sian-Jheng Lin, Wei-Ho Chung, and Yunghsiang S. Han. Novel polynomial basis and its application to Reed–Solomon erasure codes. In *IEEE 55th Annual Symposium on Foundations of Computer Science (FOCS)*, pages 316–325. IEEE, 2014.

[Lev26] Avihu Mordechai Levy. Quantum-safe Bitcoin transactions without softforks. Technical report, https://github.com/avihu28/Quantum-Safe-Bitcoin-Transactions/blob/main/paper/QSB.pdf, 2026. Source code: https://github.com/avihu28/Quantum-Safe-Bitcoin-Transactions.

[LFKN92] Carsten Lund, Lance Fortnow, Howard Karloff, and Noam Nisan. Algebraic methods for interactive proof systems. *Journal of the ACM*, 39(4):859–868, 1992.

[Lin25] Robin Linus. BitVM3: Efficient computation on Bitcoin. https://bitvm.org/bitvm3-rsa.pdf, 2025.

[LJF25] Sergio Demian Lerner, Martin Jonas, and Ariel Futoransky. ESSPI: ECDSA/Schnorr signed program input for BitVMX. arXiv preprint arXiv:2503.02772, 2025.

[Nat15] National Institute of Standards and Technology. SHA-3 standard: Permutation-based hash and extendable-output functions. FIPS PUB 202, NIST, 2015.

[OANWO20] Jack O'Connor, Jean-Philippe Aumasson, Samuel Neves, and Zooko Wilcox-O'Hearn. BLAKE3: One function, fast everywhere. https://github.com/BLAKE3-team/BLAKE3-specs, 2020.

[Pha26] Vanhger Pham. Deferred binding: Extending BABE for dynamic public inputs in GOAT BitVM3. GOAT Research, HackMD, 2026. Accessed 5 November 2026. Implementation: GOATNetwork/bitvm2-gc, branch `feat/goat-bitvm3`, commit `c23130a`.

[Plo26] Plonky3 contributors. Plonky3: binary tower fields, WHIR and multi-STARK. Source code, commit `ca4da2daa2e0dfdd31419af5c870813083398955`, 2026.

[Sho94] Peter W. Shor. Algorithms for quantum computation: Discrete logarithms and factoring. In *35th Annual Symposium on Foundations of Computer Science*, pages 124–134. IEEE, 1994.

[SRG+26] Lev Soukhanov, Yaroslav Rebenko, Muhammad El Gebali, Mikhail Komarov, Handan Kilinc-Alper, Michel Abdalla, and Patrick Towa. Implementable witness encryption from arithmetic affine determinant programs. Cryptology ePrint Archive, Paper 2026/175, 2026.

[Sta26] StarkWare. The first quantum-safe Bitcoin transaction has been mined. https://starkware.co/blog/the-first-quantum-safe-bitcoin-transaction-has-been-mined/, August 2026. Mainnet transaction: https://mempool.space/tx/305a24ffea912b9cf428f29ebf952321c96dab5bab284fc0d0801562f5abab07.

[TF26] Mukesh Tiwari and Aaron Feickert. Antichain Winternitz: Guaranteed garbled-circuit label revelation on Bitcoin with permissionless recovery. Cryptology ePrint Archive, Paper 2026/1568, 2026.

[WAA+26] Robin Linus Woll, Ioannis Alexopoulos, Lukas Aumayr, Zeta Avarikioti, Matteo Maffei, and David Tse. BitVM3: Efficient Bitcoin bridges via garbled circuits. Cryptology ePrint Archive, Paper 2026/933, 2026.

[Wie88] Doug Wiedemann. An iterated quadratic extension of GF(2). *The Fibonacci Quarterly*, 26(4):290–295, 1988.

[WNT20] Pieter Wuille, Jonas Nick, and Anthony Towns. BIP 342: Validation of Taproot scripts. Bitcoin Improvement Proposal, 2020. Defines Tapscript signature semantics and resource limits.

[ZKM26] ZKM. Ziren: a MIPS zero-knowledge virtual machine, branch `feat/binary-whir-blake3`. Source code, commit `80af2779c17d4aa40ba2cafdea21734812892b4e`, 2026.

[ZRE15] Samee Zahur, Mike Rosulek, and David Evans. Two halves make a whole: Reducing data transfer in garbled circuits using half gates. In *Advances in Cryptology – EUROCRYPT 2015, Part II*, volume 9057 of *LNCS*, pages 220–250. Springer, 2015.